Ingénierie des circuits intégrés

MASTER IESE

2025-2026

PARTIE 2

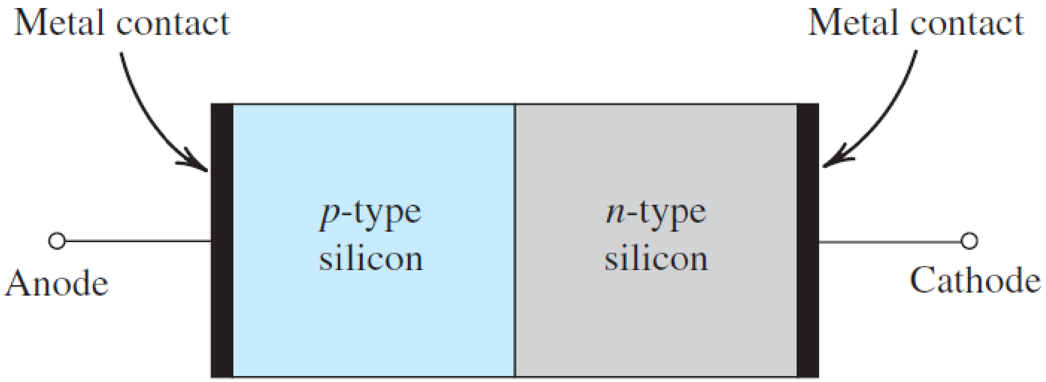
Plan

- Physique des composants et technologies des capteurs
- Phénomènes de transports dans les semiconducteurs
- Jonction : Jonction PN et Jonction métal semiconducteur
- Structure MOS, transistor MOS
- Transistors : Bipolaire et MOS

pn Junction/ p-type n-type semiconductor

Using Semiconductor

- The first practical semiconductor structure—the *pn* junction.
- The *pn* junction implements the diode (electronic component) and plays the dominant role in the structure and operation of the bipolar junction transistor (BJT, electronic component).
- The *pn* junctions is very important to the study of the MOSFET operation

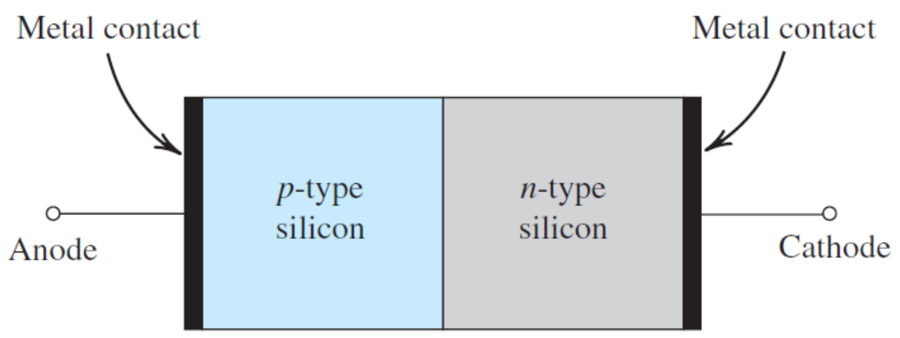


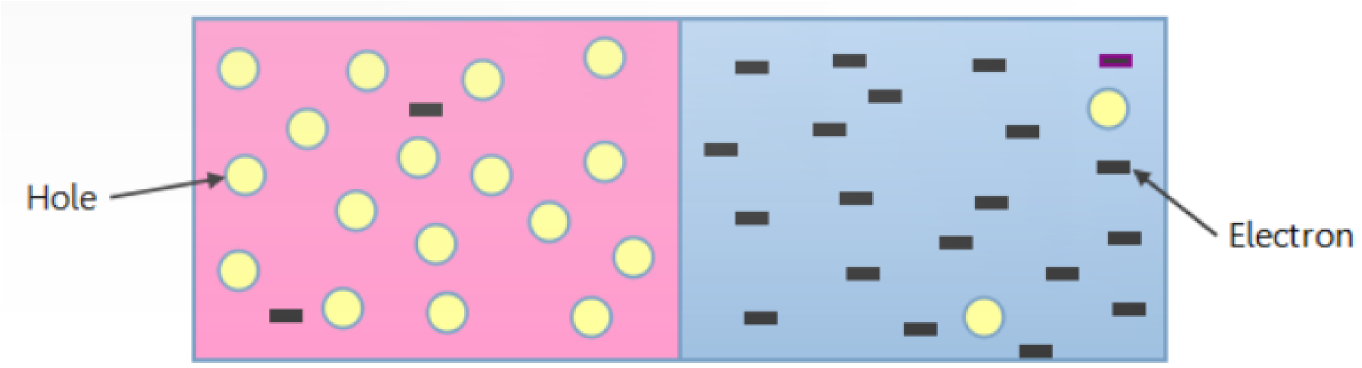
Simplified physical structure of the *pn* junction. As the *pn* junction implements the junction diode, its terminals are labeled anode and cathode.

pn Junction

Physic structure

- It consists of a p-type semiconductor (e.g., silicon) brought into close contact with an n-type semiconductor material (also silicon).
- In actual practice, both the p and n regions are part of the same silicon crystal;
- The pn junction is formed within a single silicon crystal by creating regions of different dopings (p and n regions).

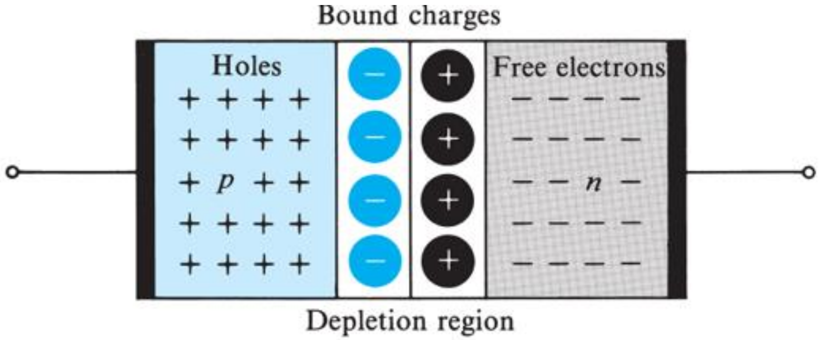




Situation before formation of depletion region

pn Junction

Operation with Open-Circuit Terminals



Step 1: Massive Diffusion

Holes from the p-side (where they are majority carriers) diffuse into the n-side.

Electrons from the n-side (where they are majority carriers) diffuse into the p-side.

Step 2: Ionization and Charge Buildup

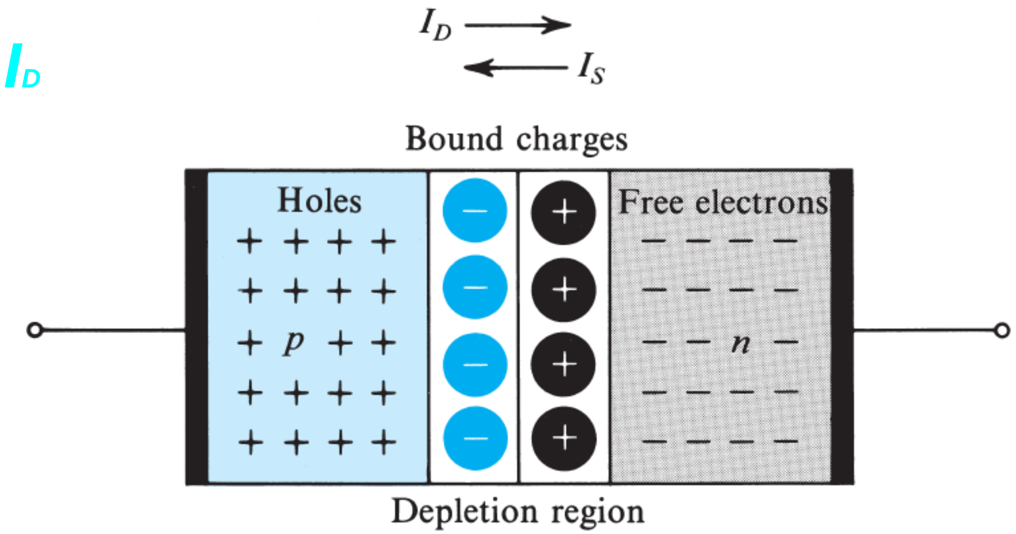
When a hole diffuses into the n-side, it leaves behind a negatively charged acceptor atom (N_a^-) in the p-side.

When an electron diffuses into the p-side, it leaves behind a positively charged donor atom (N_d^+) in the n-side.

This creates a region on either side of the junction that is depleted of mobile charge carriers (holes and electrons), hence the name Depletion Region or Space Charge Region.

pn Junction

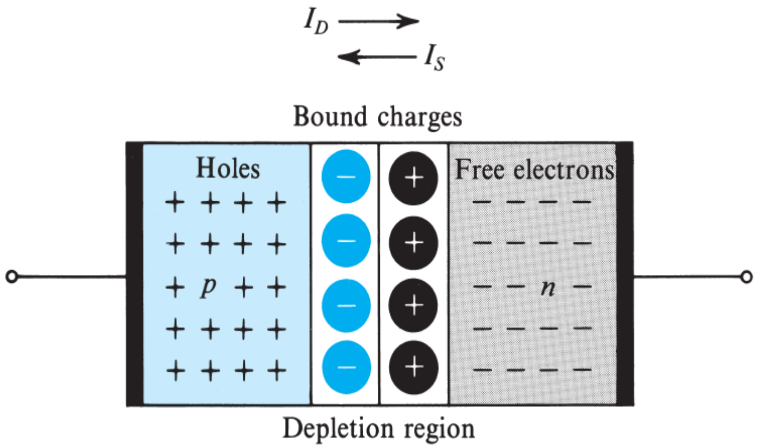
The Diffusion Current I_D



- Because the concentration of holes is high in the *p* region and low in the *n* region, holes diffuse across the junction from the *p* side to the *n* side. Similarly, electrons diffuse across the junction from the *n* side to the *p* side.
- These two current components add together to form the diffusion current I_D, whose direction is from the *p* side to the *n* side

pn Junction

The Drift Current I_S and Equilibrium



In addition to the current component I_D due to majority-carrier diffusion, a component due to minority-carrier drift exists across the junction.

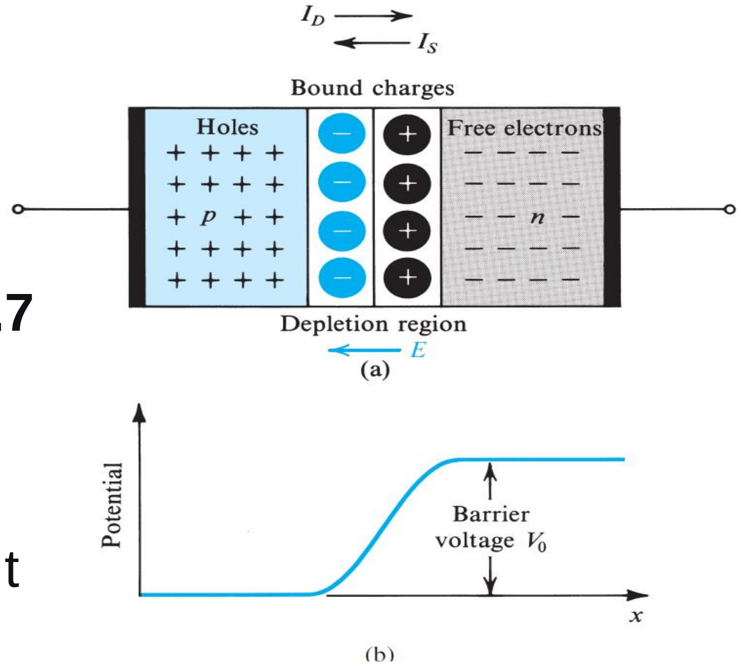
Specifically, some of the thermally generated holes in the *n* material move toward the junction and reach the edge of the depletion region. There, they experience the electric field in the depletion region, which sweeps them across that region into the *p* side.

Similarly, some of the minority thermally generated electrons in the *p* material move to the edge of the depletion region and get swept by the electric field in the depletion region across that region into the *n* side.

These two current components—electrons moved by drift from *p* to *n* and holes moved by drift from *n* to *p*—add together to form the drift current I_S, whose direction is from the *n* side to the *p* side of the junction, as indicated in Fig

pn Junction

The Drift Current I_S and Equilibrium



Step 3: Built-in Electric Field and Potential

•The fixed negative charges on the p-side and positive charges on the n-side create an **internal electric field (E)** across the junction.

•This field creates a **built-in potential (V_0 or $V_{\beta i}$)**, typically around **0.6 to 0.7 V for Silicon** at room temperature.

Step 4: Equilibrium

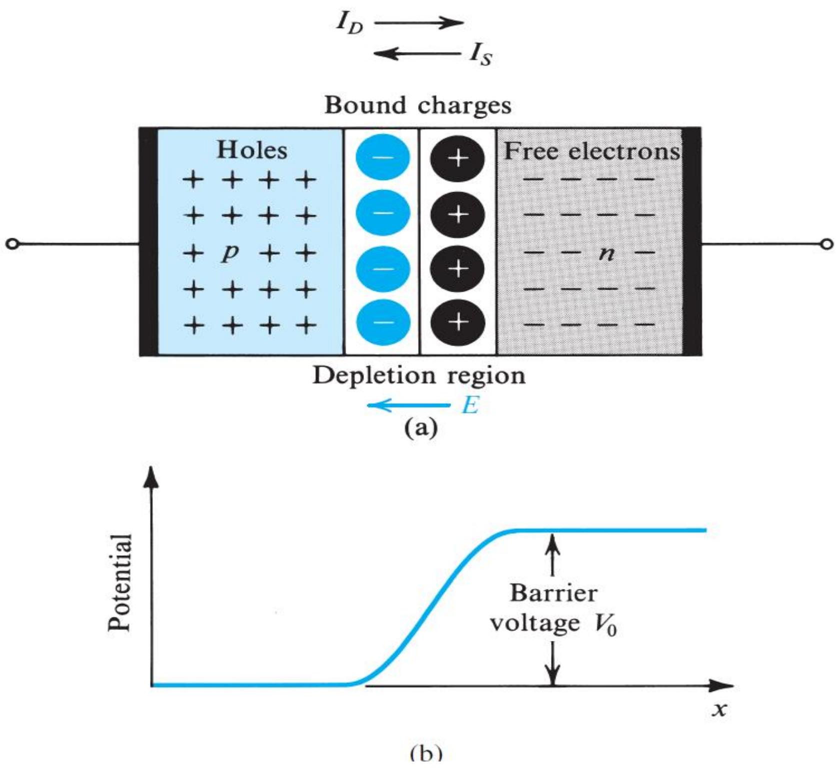
•The built-in electric field opposes the further diffusion of majority carriers. It now acts as a "barrier."

•A state of dynamic equilibrium is reached where the tendency for carriers to diffuse is exactly balanced by the electric field pushing them back.

pn Junction

The Drift Current *IS* and Equilibrium

Under open-circuit conditions no external current exists; thus the two opposite currents across the junction must be equal in magnitude: $I_D = I_s$



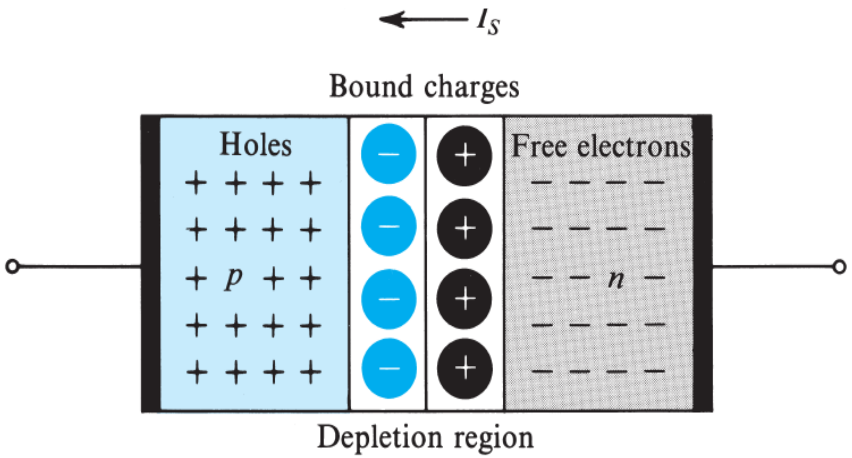
The *pn* junction with no applied voltage (open-circuited terminals). **(b)** The potential distribution along an axis perpendicular to the junction

pn Junction

The Junction Built-in Voltage

With no external voltage applied, the barrier voltage V_0 across the *pn* junction can be shown to be given by

$$V_0 = V_T \ln\left(\frac{N_A N_D}{n_i^2}\right)$$

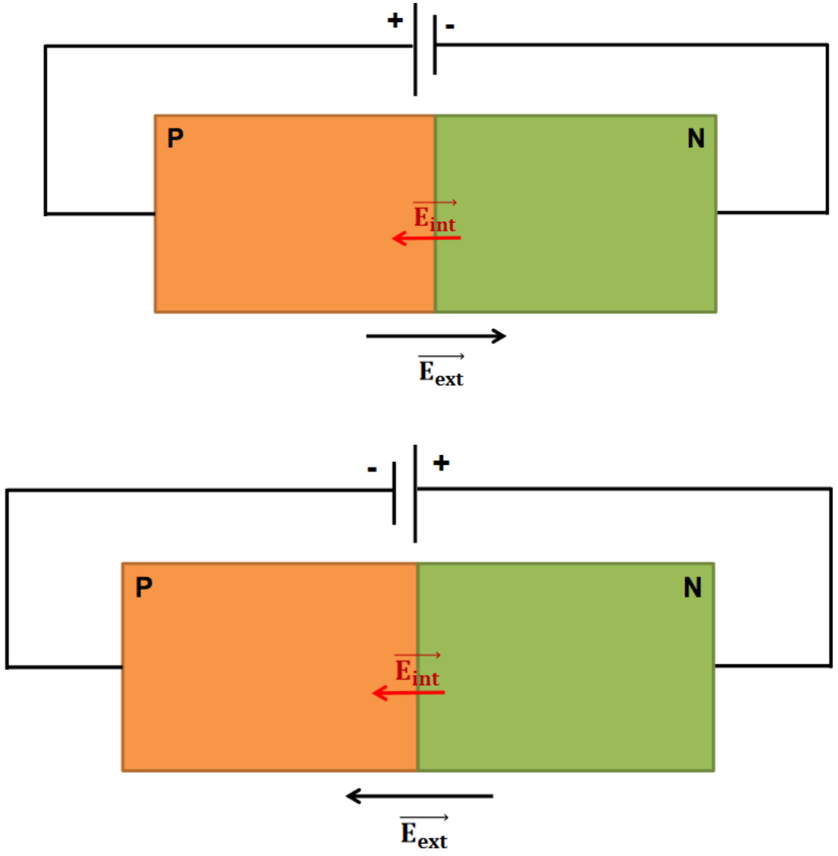


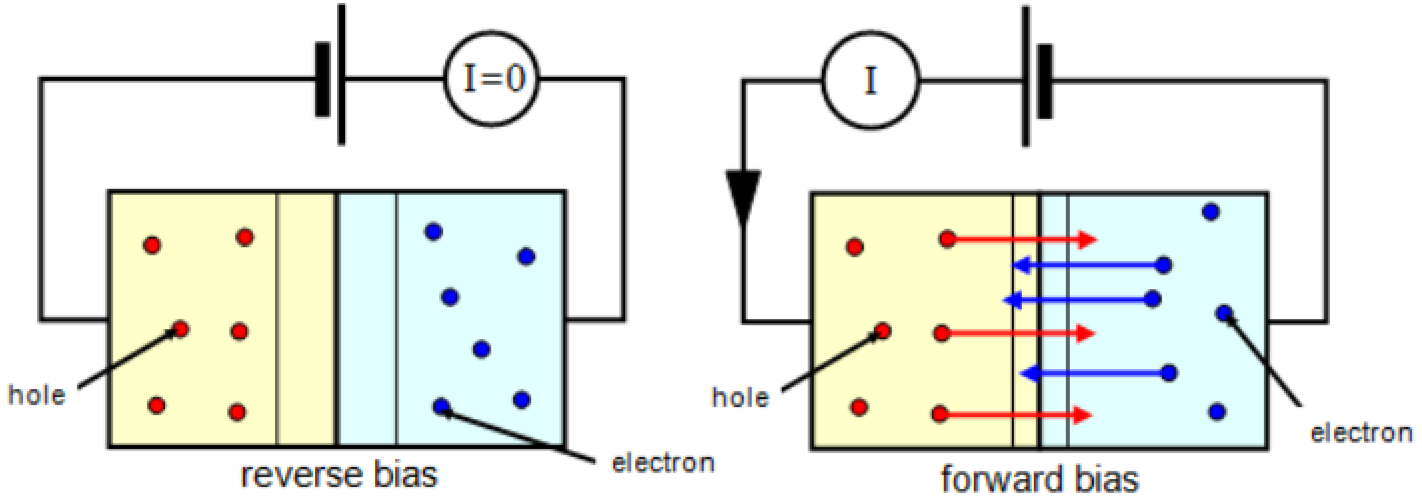
where N_A and N_D are the doping concentrations of the *p* side and *n* side of the junction, respectively.

pn Junction with an Applied Voltage

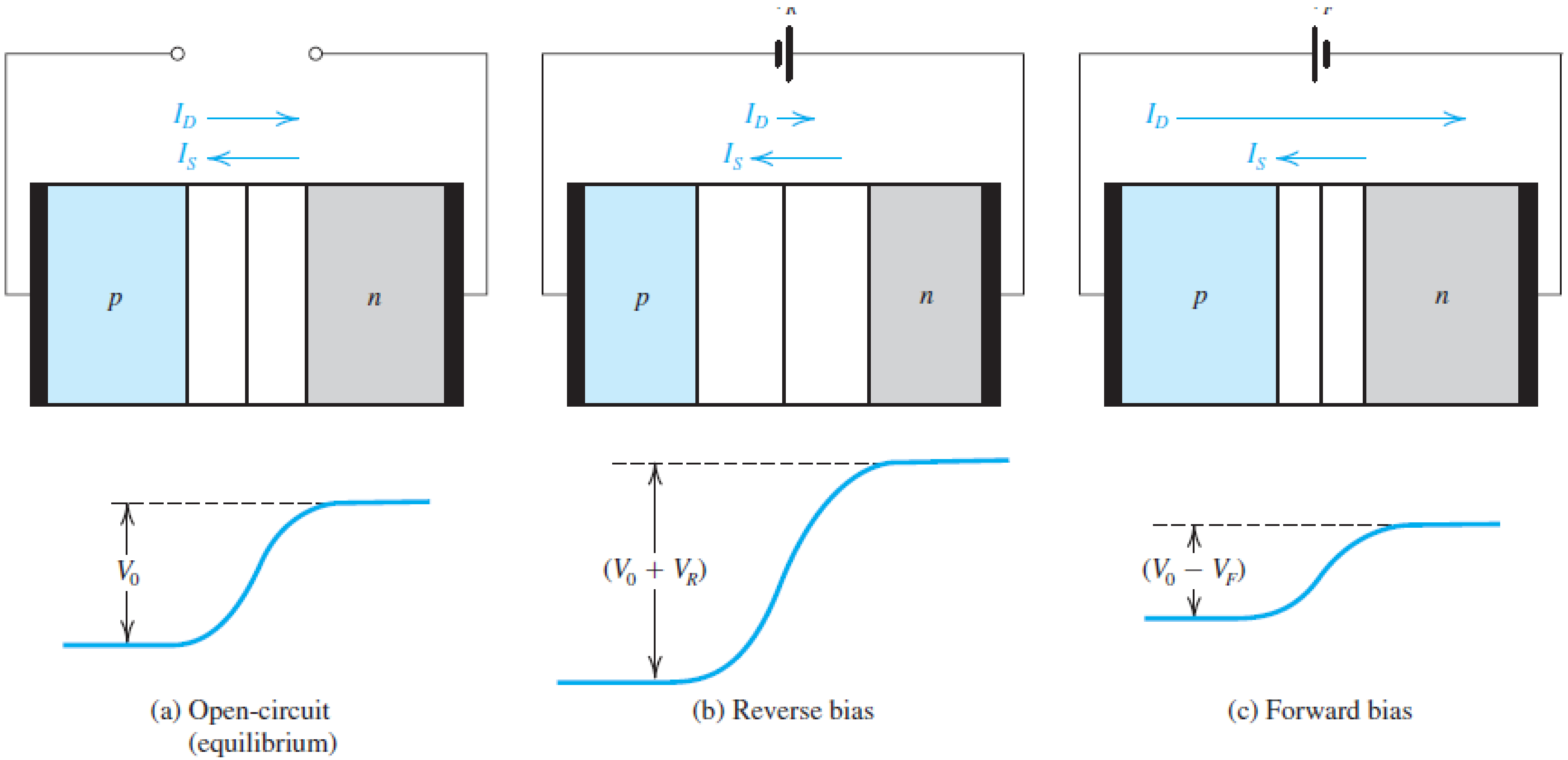
The Junction Built-in Voltage

- If the voltage is applied so that the *p* side is made more positive than the *n* side, it is referred to as a forward-bias voltage “**polarisation en direct**”.
- If our applied dc voltage is such that it makes the *n* side more positive than the *p* side, it is said to be a reverse-bias voltage “**polarisation en inverse**”.





pn Junction with an Applied Voltage



The *pn* junction in: **(a)** equilibrium; **(b)** reverse bias; **(c)** forward bias.

pn Junction with an Applied Voltage

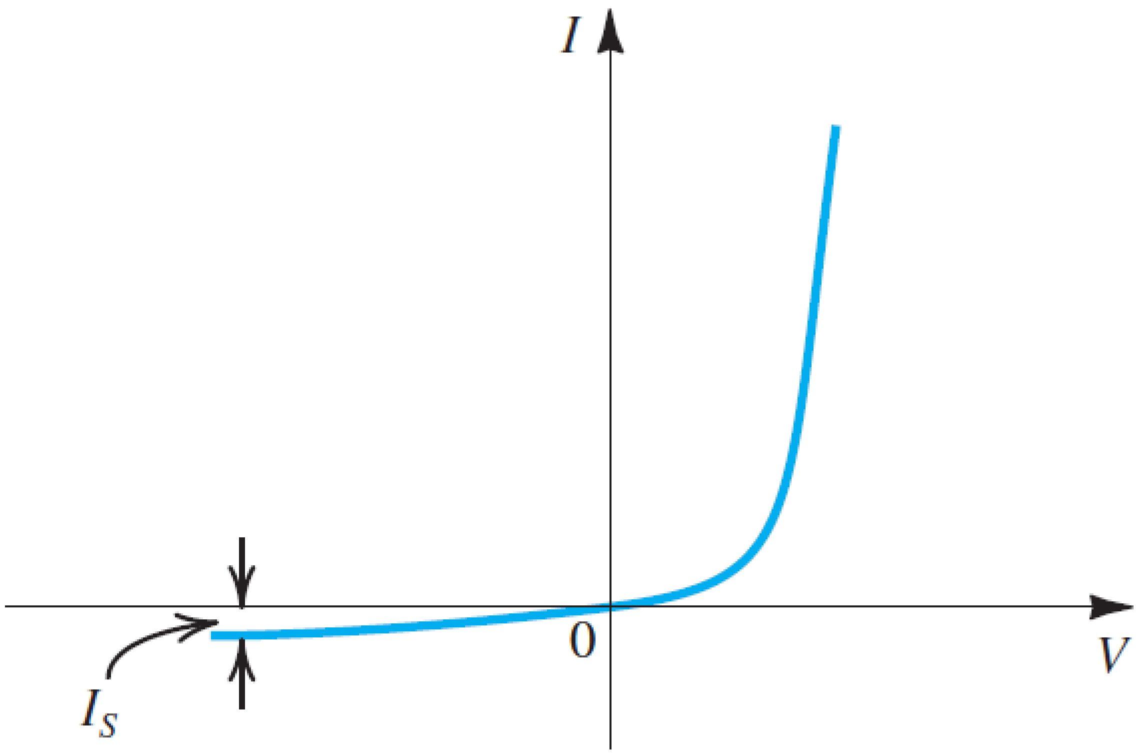
The Current–Voltage Relationship of the Junction

The total current of the junction is given by :

$$I = I_S\left(e^{V/V_T} - 1\right)$$

Where V note the voltage applied on the pn junction and

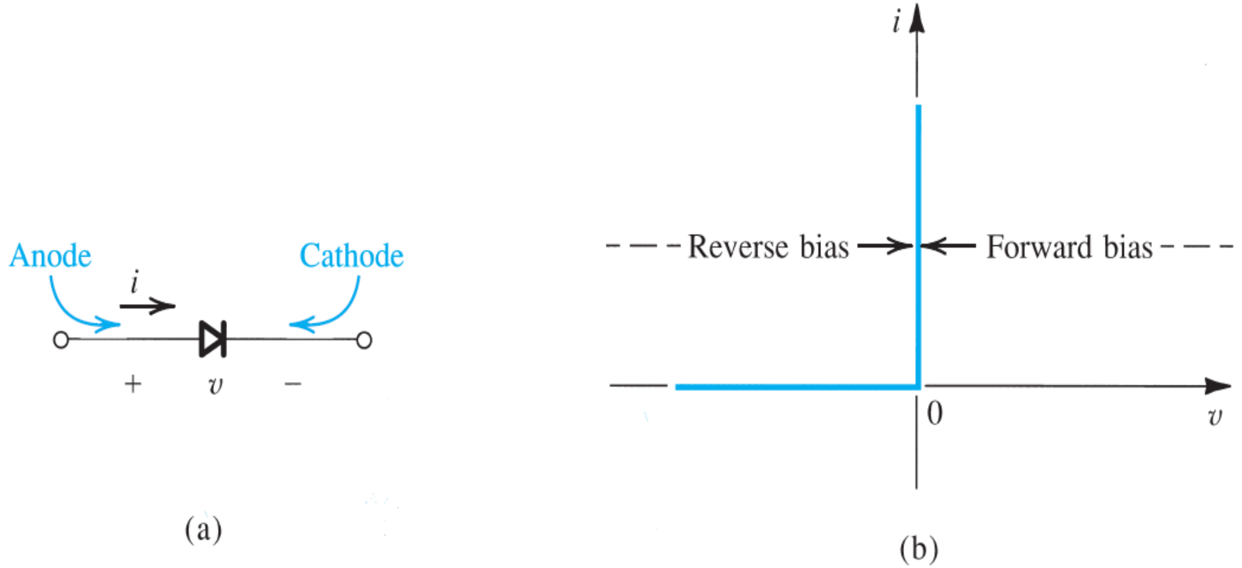
V_T is the thermal voltage (~26 V)

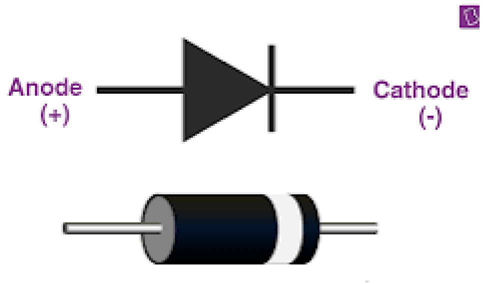


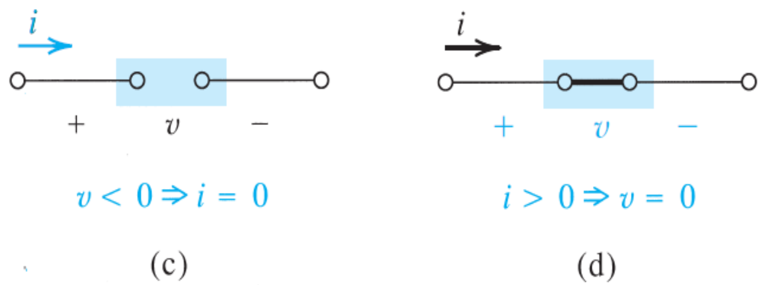
The *pn* junction *I*–*V* characteristic.

Diode

Current–Voltage Characteristic of Ideal Diode





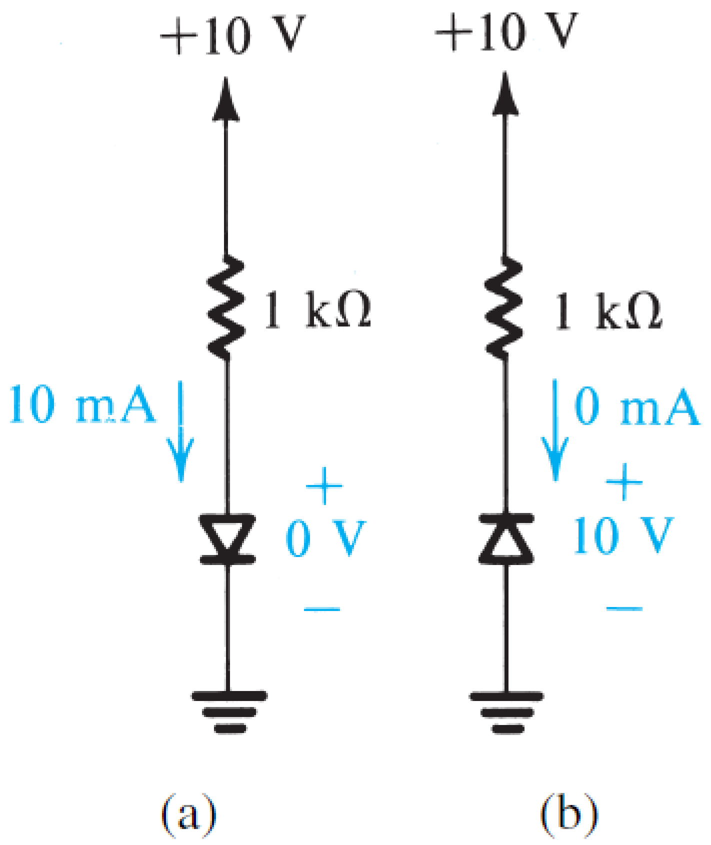


The ideal diode: **(a)** diode circuit symbol; **(b)** *i–v* characteristic; **(c)** equivalent circuit in the reverse direction; **(d)** equivalent circuit in the forward direction.

Diode

Sample of Ideal Diode application

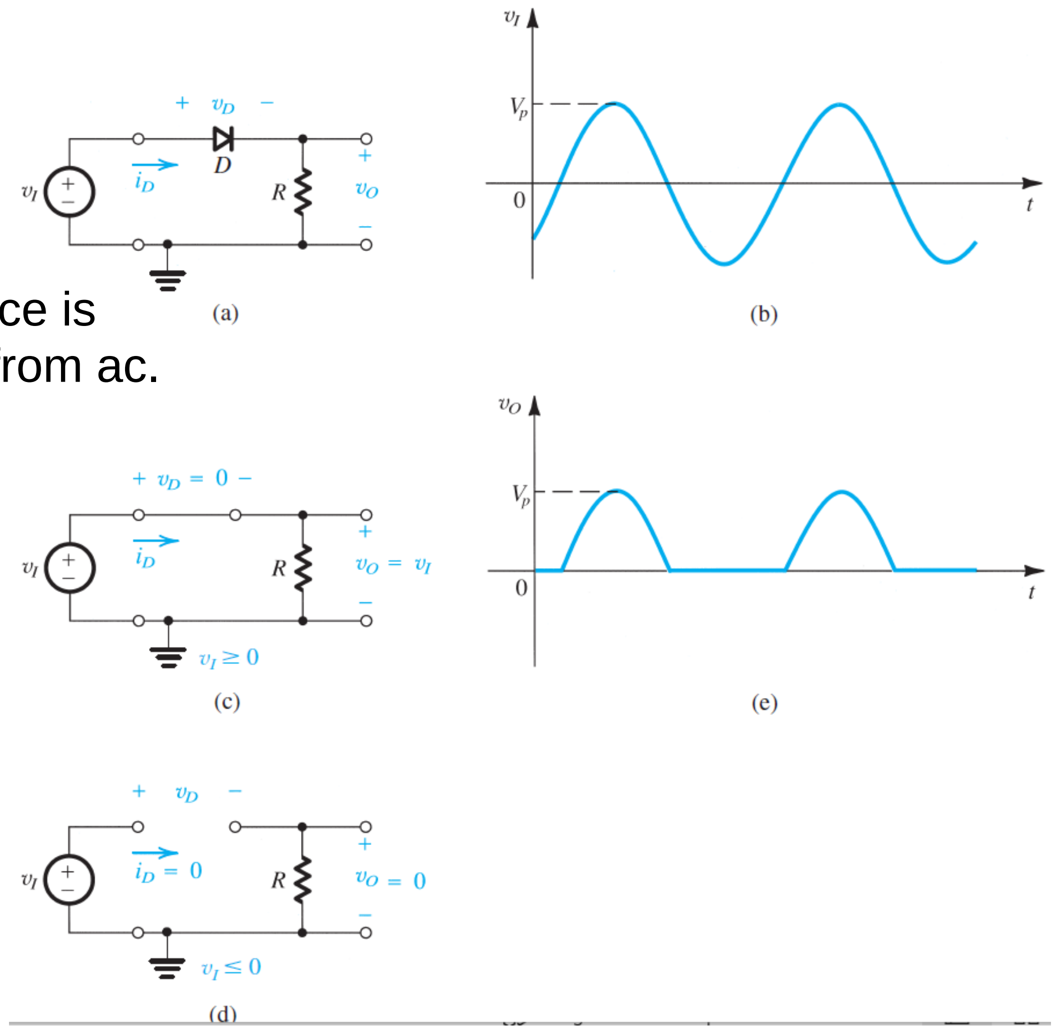
The two modes of operation of ideal diodes and the use of an external circuit to limit **(a)** the forward current and **(b)** the reverse voltage.



Diode Application

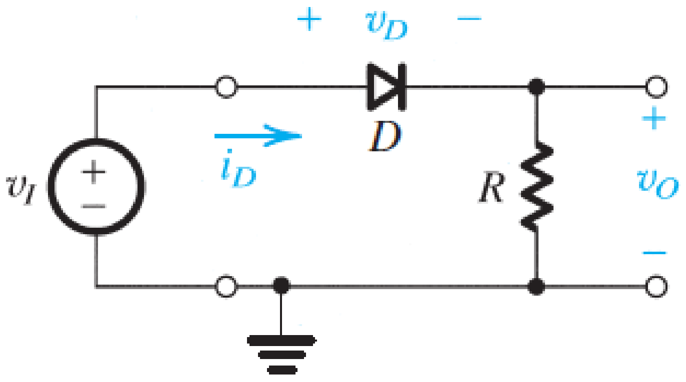
A Simple Application: The Rectifier

the circuit of Fig. (a) **rectifies** the signal and hence is called a **rectifier**. It can be used to generate dc from ac.



Diode Application

A Simple Application: The Rectifier



(a)

Exercise

1) For the circuit in Fig. (a), sketch the transfer characteristic v_O versus v_I.

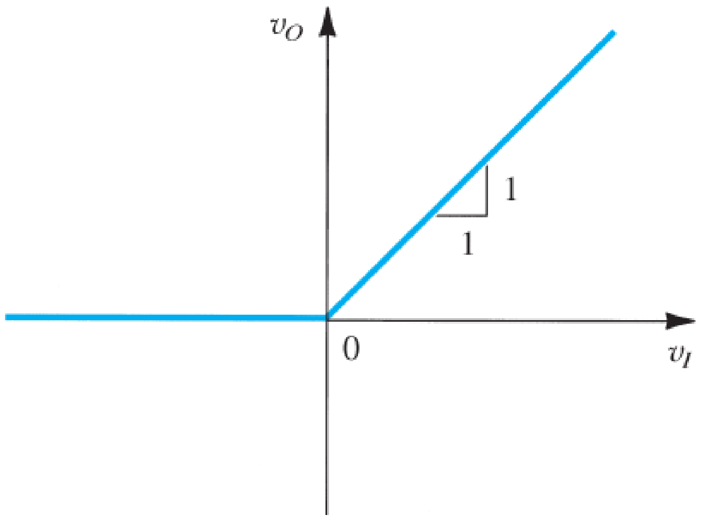
2) For the circuit in Fig. (a), sketch the waveform of v_D.

Diode Application

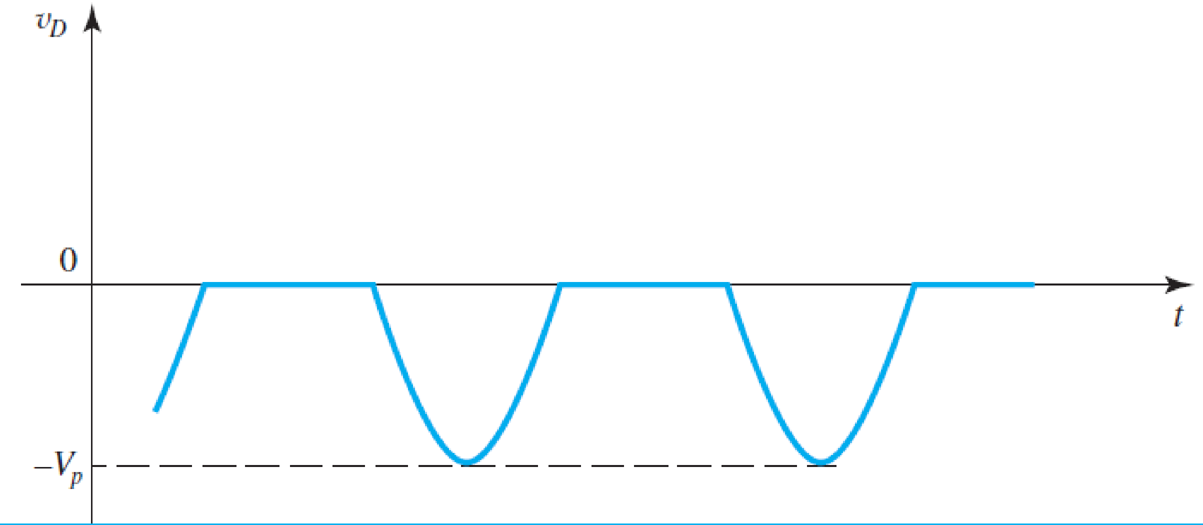
A Simple Application: The Rectifie

Exercise/Ans

1) v_o (v_I.)



2) $v_D = v_I - v_O$

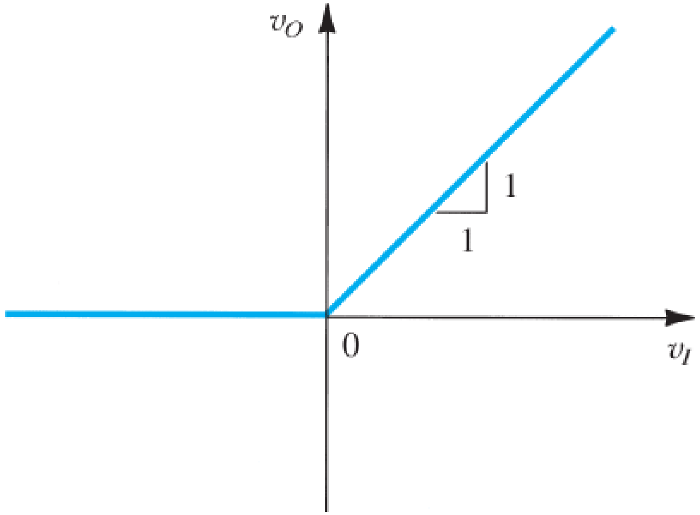


Diode Application

A Simple Application: The Rectifie

Exercise/Ans

1) v_o (v_I .)



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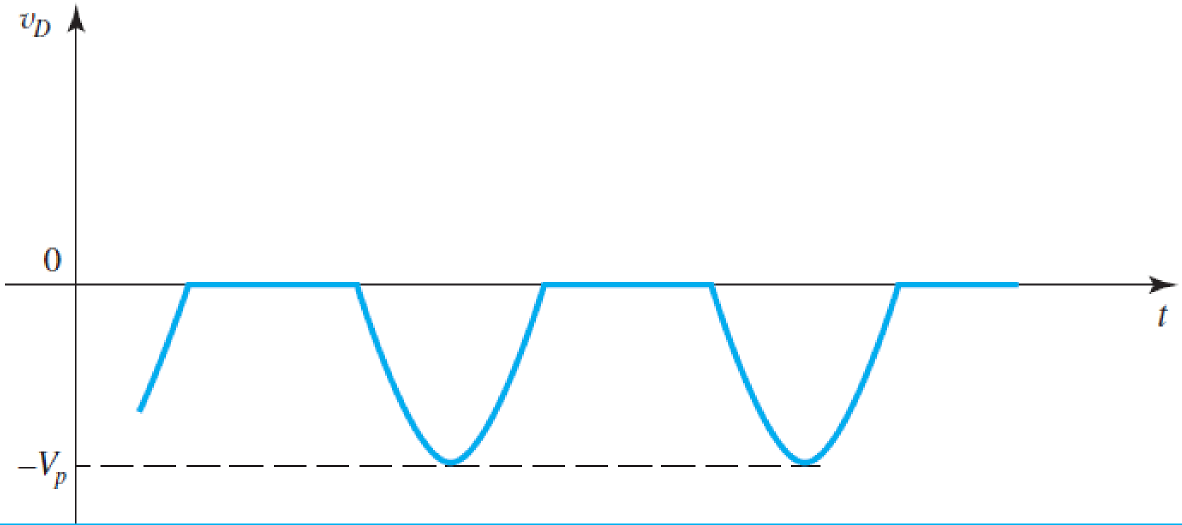


Transistor MOSFET

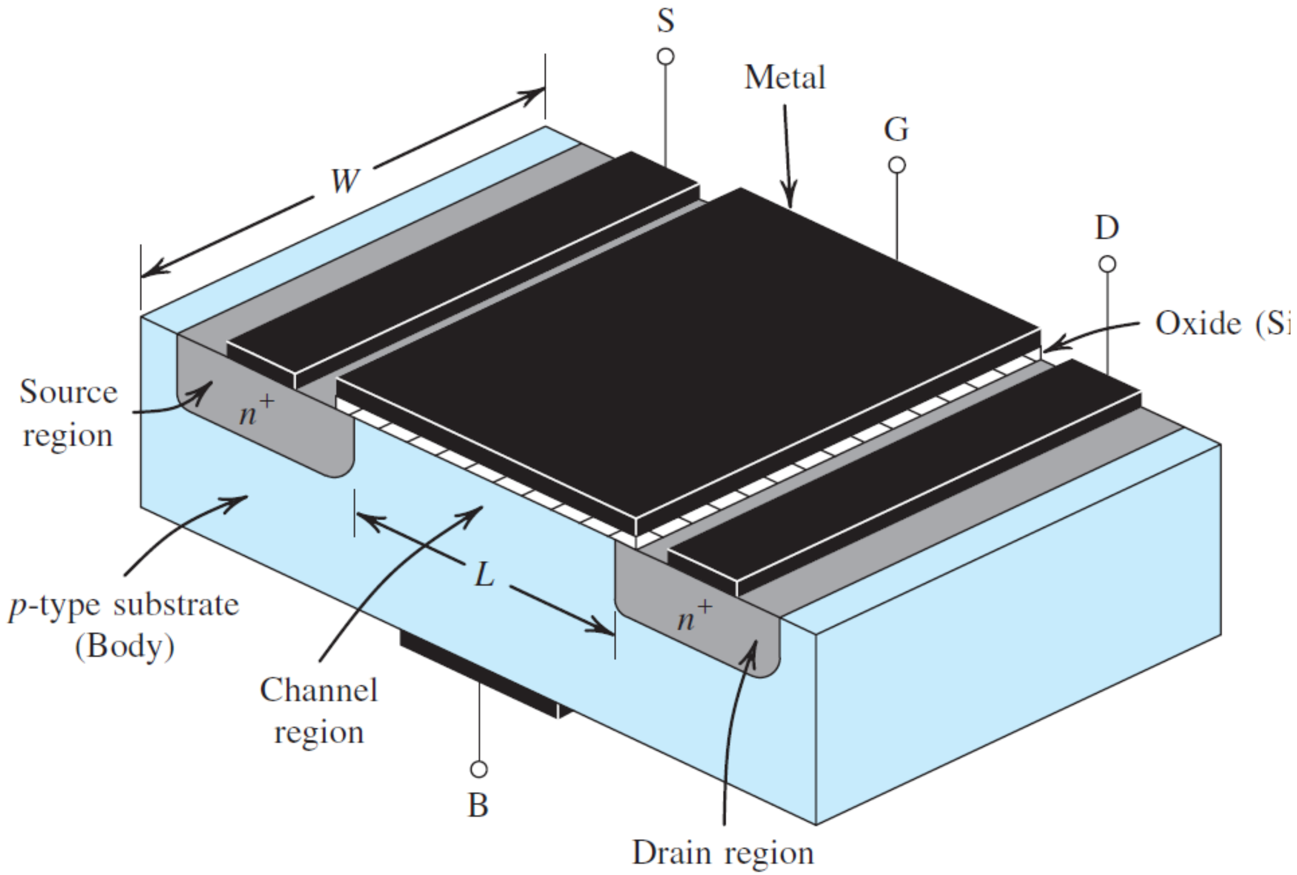
THE FIRST FIELD-EFFECT DEVICES:

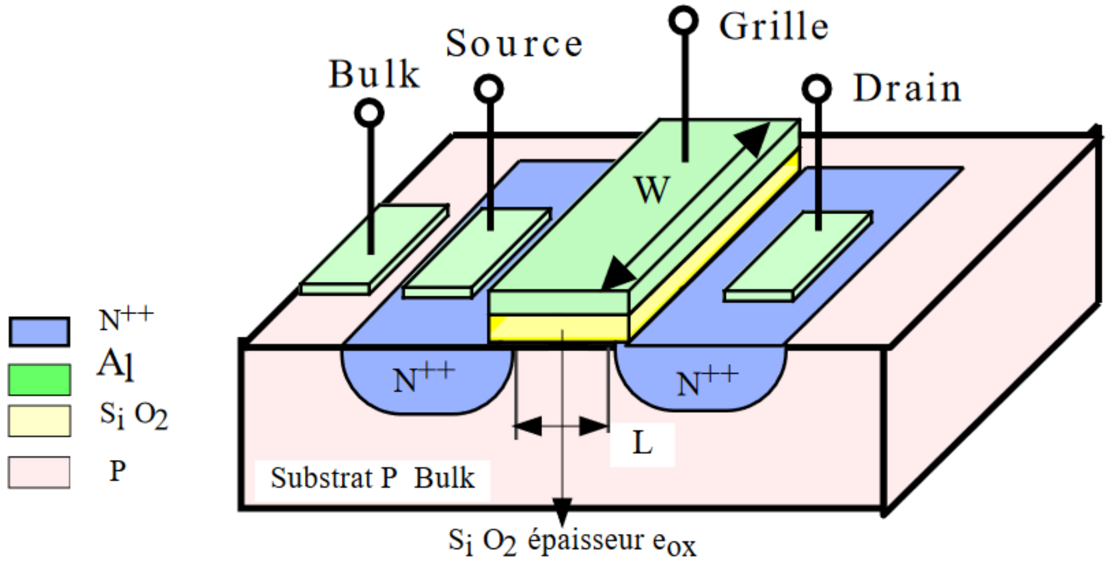
In 1925 a patent for solid-state electric-field-controlled conductor was filed in Canada by Julius E. Lilienfeld, a physicist at the University of Leipzig, Germany. Other patent refinements followed in the United States in 1926 and 1928. Regrettably, no research papers were published. Consequently, in 1934 Oskar Heil, a German physicist working at the University of Cambridge, U.K., filed a patent on a similar idea. But all these early concepts of electric-field control of a semiconducting path languished because suitable technology was not available.

The invention of the bipolar transistor in 1947 at Bell Telephone Laboratories resulted in the speedy development of bipolar devices, a circumstance that further delayed the development of field-effect transistors. Although the field-effect device was described in a paper by William Shockley in 1952, it was not until 1960 that a patent on an insulated-gate field-effect device, the MOSFET, was filed by Dawon Kahng and Martin Atalla, also at Bell Labs. Clearly, the idea of field-effect control for amplification and switching has changed the world. With integrated-circuit chips today containing billions of MOS devices, MOS dominates the electronics world!

Transistor MOSFET/Device Structure and Physical Operation

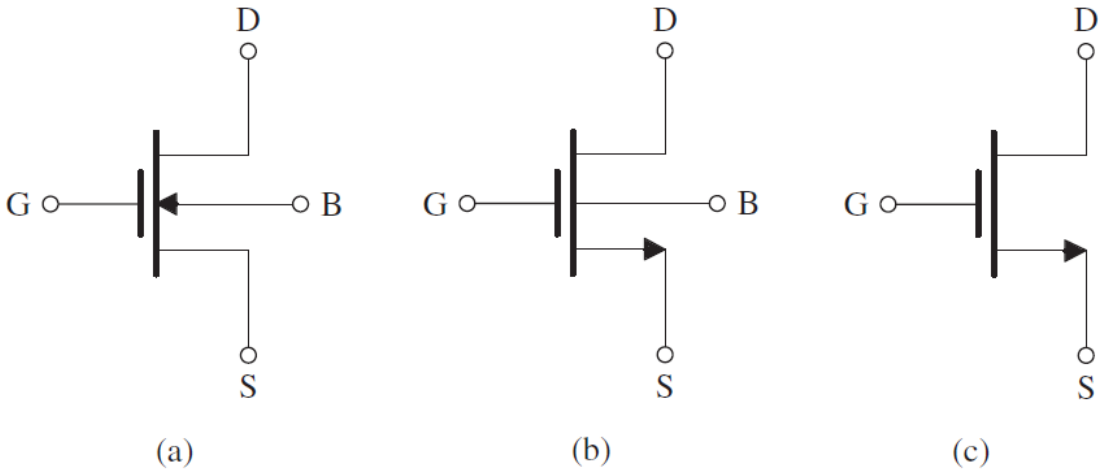
Device Structure





Transistor MOSFET/Device Structure and Physical Operation

Device Symbol



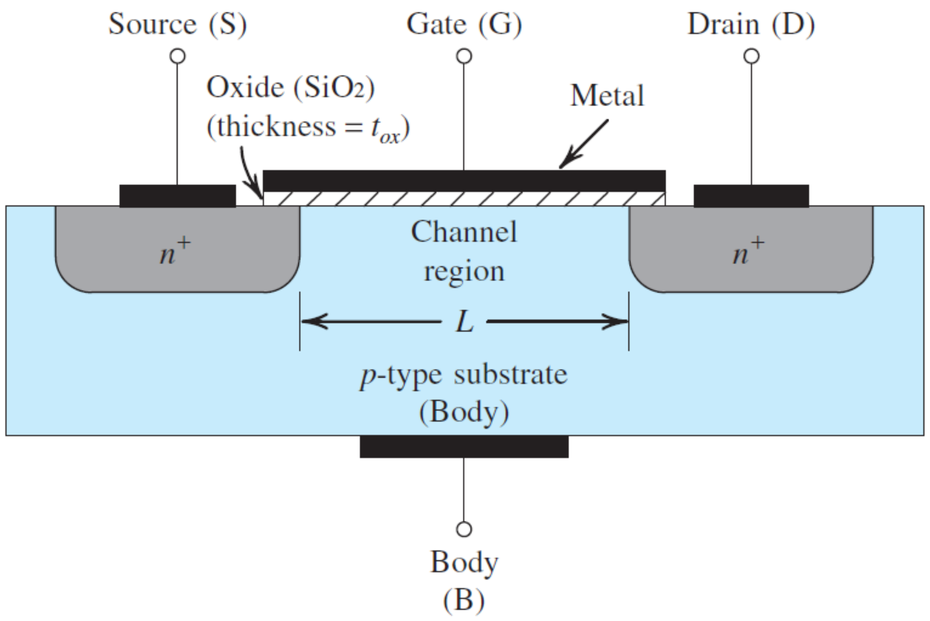
(a) Circuit symbol for the *n*-channel enhancement-type MOSFET. **(b)** Modified circuit symbol with an arrowhead on the source terminal to distinguish it from the drain and to indicate device polarity (i.e., *n* channel). **(c)** Simplified circuit symbol to be used when the source is connected to the body or when the effect of the body on device operation is unimportant.

Transistor MOSFET/Device Structure and Physical Operation

Operation with Zero Gate Voltage

With zero voltage applied to the gate, two back-to-back diodes exist in series between drain and source. One diode is formed by the *pn* junction between the n_+ drain region and the *p*-type substrate, and the other diode is formed by the *pn* junction between the *p*-type substrate and the n_+ source region.

These back-to-back diodes prevent current conduction from drain to source when a voltage v_{DS} is applied. In fact, the path between drain and source has a very high resistance (of the order of $10^{12}\ \Omega$).

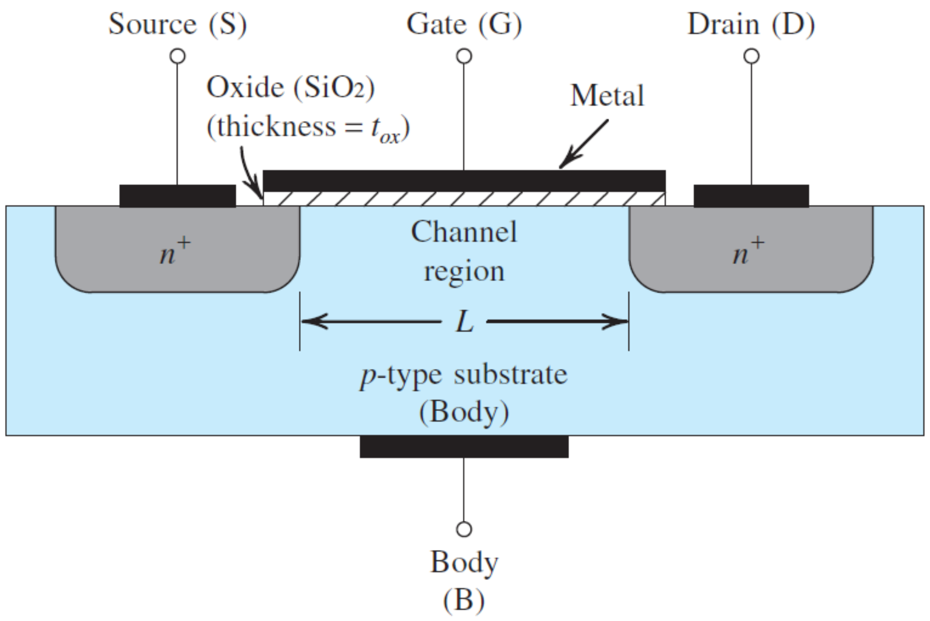


Cross section. Typically L = 0.03 μm to 1 μm, W = 0.05 μm to 100 μm, and the thickness of the oxide layer (*tox*) is in the range of 1 to 10 nm.

Transistor MOSFET/Device Structure and Physical Operation

Operation with Zero Gate Voltage

- Observe that the substrate forms pn junctions with the source and drain regions.
- In normal operation these pn junctions are kept reverse biased at all times.
- Since, the drain will always be at a positive voltage relative to the source, the two pn junctions can be effectively cut off by simply connecting the substrate terminal to the source terminal.



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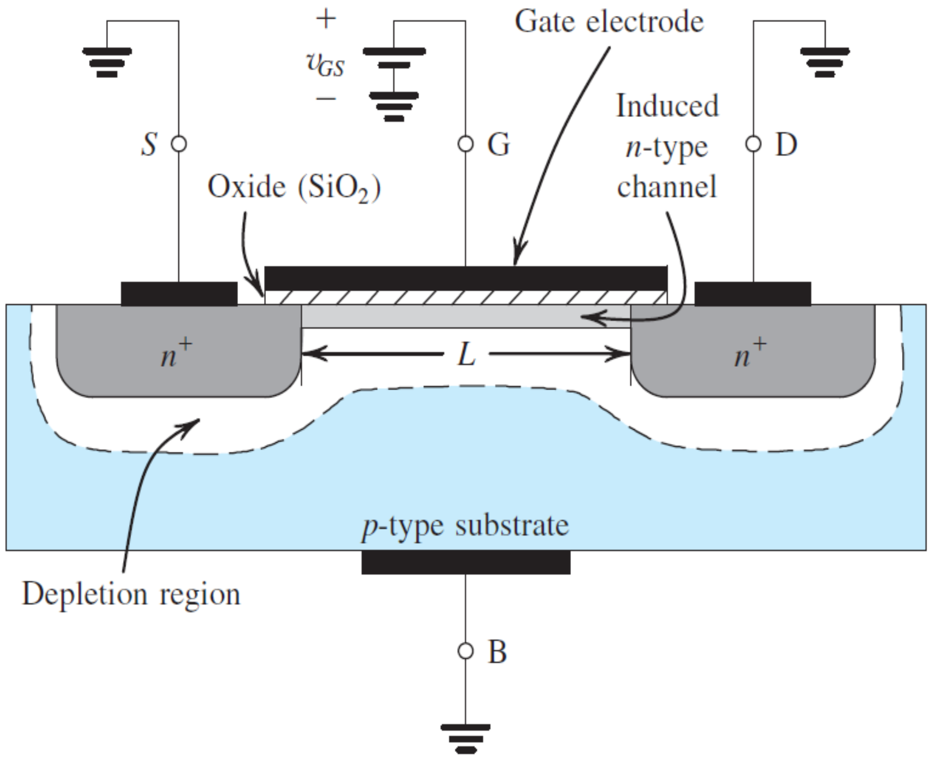
Transistor MOSFET/Device Structure and Physical Operation

Creating a Channel for Current Flow/Gate Voltage positive

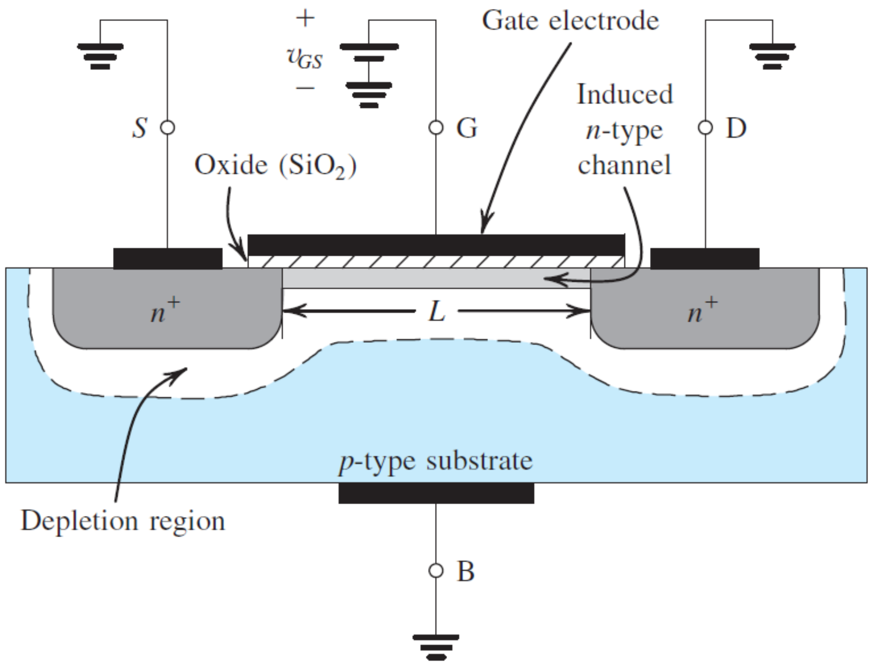
- grounded the source and the drain and applied a positive voltage to the gate. Since the source is grounded, the gate voltage appears in effect between gate and source and thus is denoted v_{GS}.
- The positive voltage on the gate causes, in the first instance, the free holes (which are positively charged) to be repelled from the region of the substrate under the gate (the channel region).
- These holes are pushed downward into the substrate, leaving behind a carrier-depletion region. The depletion region is populated by the bound negative charge associated with the acceptor atoms.
- These charges are “uncovered” because the neutralizing holes have been pushed downward into the substrate.



Transistor MOSFET/Device Structure and Physical Operation

Creating a Channel for Current Flow/Gate Voltage positive

- The enhancement-type NMOS transistor with a positive voltage applied to the gate. An *n* channel is induced at the top of the substrate beneath the gate.
- The voltage across this parallel-plate capacitor, that is, the voltage across the oxide, must exceed V_t for a channel to form.
- When v_{DS}=0, the voltage at every point along the channel is zero, and the voltage across the oxide (i.e., between the gate and the points along the channel) is uniform and equal to v_{GS}.
- The excess of v_{GS} over V_t is termed the **effective voltage** or the **overdrive voltage** and is the quantity that determines the charge in the channel.



The value of v_{GS} at which a sufficient number of mobile electrons accumulate in the channel region to form a conducting channel is called the **threshold voltage** and is denoted V_t

Transistor MOSFET/Device Structure and Physical Operation

Creating a Channel for Current Flow/ Gate Voltage positive

- We can express the magnitude of the electron charge in the channel by

$$|Q| = C_{ox}(WL)v_{OV}$$

where C_{ox}, called the **oxide capacitance**, is the capacitance of the parallel-plate capacitor per unit gate area (in units of F/m^2), W is the width of the channel, and L is the length of the channel. The oxide capacitance C_{ox} is given by

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$$

where ϵ_{ox} is the permittivity of the silicon dioxide,

$$\epsilon_{ox} = 3.9\epsilon_0 = 3.9 \times 8.854 \times 10^{-12} = 3.45 \times 10^{-11} \text{ F/m}$$

Transistor MOSFET/Device Structure and Physical Operation

Creating a Channel for Current Flow/ Gate Voltage positive

The oxide thickness t_{ox} is determined by the process technology used to fabricate the MOSFET.

As an example, for a process with t_{ox}= 4 nm,

$$C_{ox} = \frac{3.45 \times 10^{-11}}{4 \times 10^{-9}} = 8.6 \times 10^{-3}\ \mathrm{F/m^2} = 8.6\ \mathrm{fF/\mu m^2}$$

For a MOSFET fabricated in this technology with a channel length L = 0.18 μm and a channel width W = 0.72 μm, the total capacitance between gate and channel is $C = C_{ox}WL$ = 8.6×0.18×0.72 = 1.1 fF

Transistor MOSFET/Device Structure and Physical Operation

Creating a Channel for Current Flow/ Gate Voltage positive and small Vds

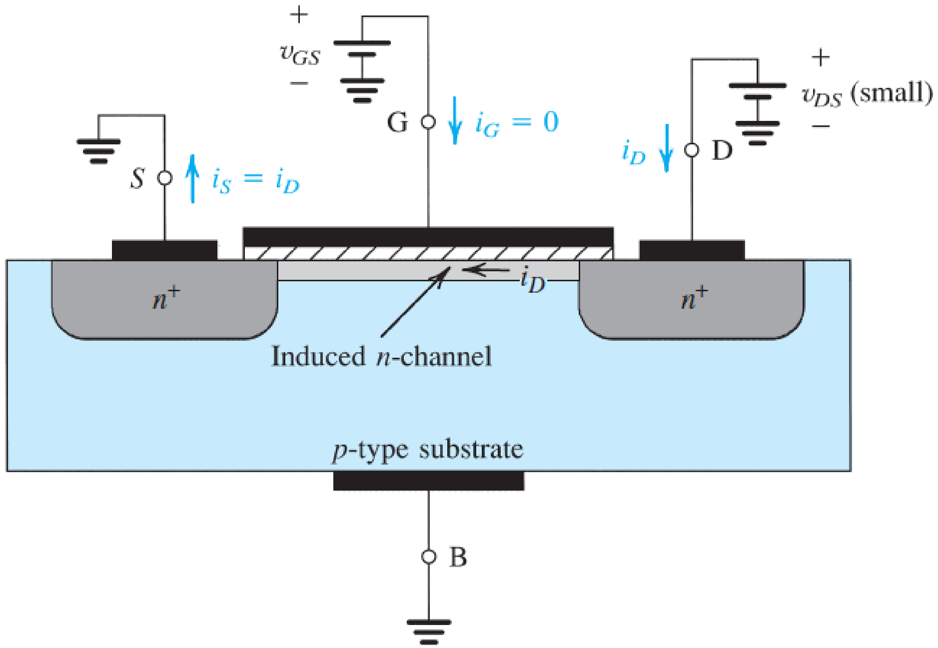
- Having induced a channel, and now if a positive voltage v_{DS} is applied between drain and source.

We first consider the case where v_{DS} is small (i.e., 50 mV).

- The voltage v_{DS} causes a current i_D to flow through the induced *n* channel.
- Current is carried by free electrons traveling from source to drain.
- The current in the channel, i_D, will be from drain to source,

$$i_D = \left[(\mu_n C_{ox})\left(\frac{W}{L}\right)v_{OV}\right]v_{DS} = \left[(\mu_n C_{ox})\left(\frac{W}{L}\right)(v_{GS} - V_t)\right]v_{DS}$$

- where μn is the mobility of the electrons at the surface of the channel and V_{ov} denote the overdrive voltage ($V_{ov}=V_{GS}-V_t$)



An NMOS transistor with $v_{GS} > V_t$ and with a small v_{DS} applied. The device acts as a resistance whose value is determined by v_{GS}.

Transistor MOSFET/Device Structure and Physical Operation

MOSFET as a Resistor /Gate Voltage positive and small Vds

Thus, for small v_{DS}, the channel behaves as a linear resistance whose value is controlled by the overdrive voltage v_{OV} , which in turn is determined by v_{GS}:

$$i_D = \left[(\mu_n C_{ox})\left(\frac{W}{L}\right)(v_{GS} - V_t)\right] v_{DS}$$

The conductance g_{DS} of the channel can be found from Eq. above as

$$g_{DS} = (\mu_n C_{ox})\left(\frac{W}{L}\right) v_{OV}$$

or

$$g_{DS} = (\mu_n C_{ox})\left(\frac{W}{L}\right)(v_{GS} - V_t)$$

Transistor MOSFET/Device Structure and Physical Operation

MOSFET as a Resistor /Gate Voltage positive and small Vds

The **MOSFET transconductance parameter** k_n,

$$k_n = k_n'(W/L)$$

With

$$k_n' = \mu_n C_{ox} \ \ (\mathrm{A/V^2})$$

with v_{DS} kept small, the MOSFET behaves as a linear resistance r_{DS} whose value is controlled by the gate voltage v_{GS},

$$r_{DS} = \frac{1}{g_{DS}}$$

$$r_{DS} = \frac{1}{(\mu_n C_{ox})(W/L)v_{OV}}$$

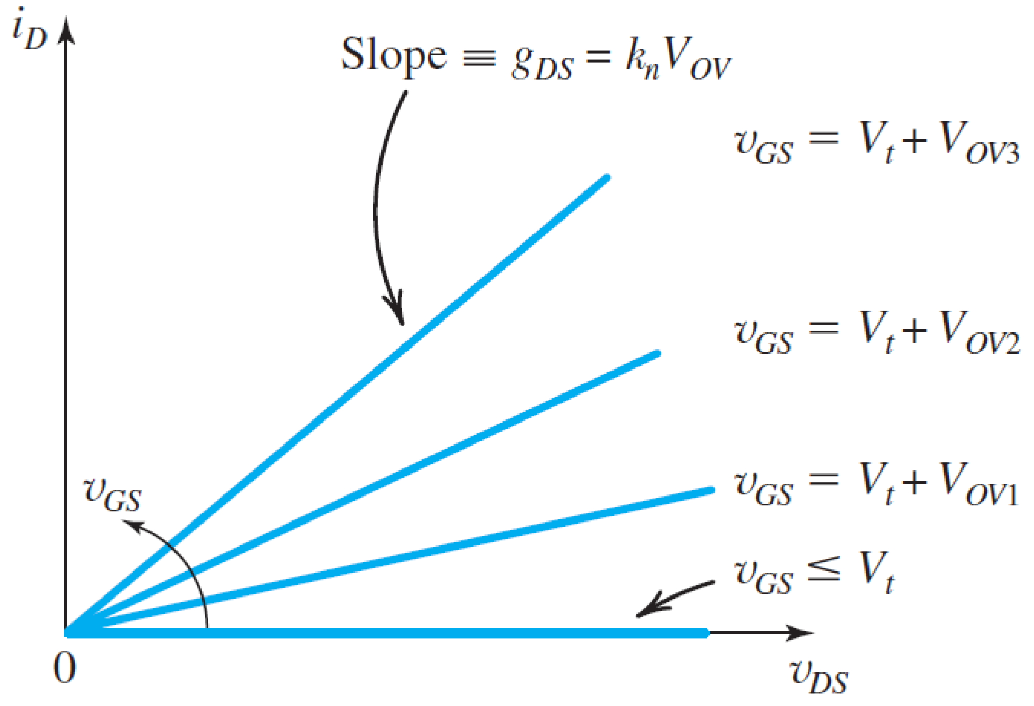
$$r_{DS} = \frac{1}{(\mu_n C_{ox})(W/L)(v_{GS} - V_t)}$$

Transistor MOSFET/Device Structure and Physical Operation

MOSFET as a Resistor /Gate Voltage positive and small Vds

The operation of the MOSFET as a voltage-controlled resistance is further illustrated in Fig. below, which is a sketch of i_D versus v_{DS} for various values of v_{GS}.

$$r_{DS} = \frac{1}{(\mu_n C_{ox})(W/L)(v_{GS} - V_t)}$$



i_D–v_{DS} characteristics of the MOSFET v_{DS}, is kept small.

Transistor MOSFET/Device Structure and Physical Operation

Example

A 0.18-μm fabrication process is specified to have t_{ox}= 4 nm, μ_n= 450 $cm^2/V \cdot s$, and V_t= 0.5 V.

1) Find the value of the process transconductance parameter k_n'. For a MOSFET with minimum length fabricated in this process,

2) Find the required value of W so that the device exhibits a channel resistance r_{DS} of 1 k at v_{GS}= 1 V.

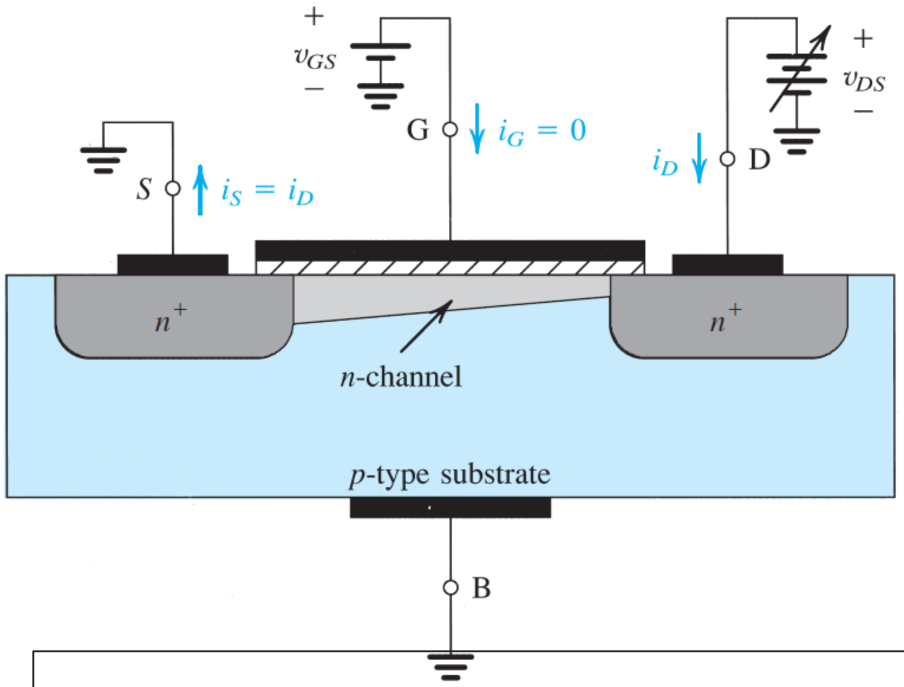
Ans.

k_n'=388 μA/V^2; W=0.93 μm

Transistor MOSFET/Device Structure and Physical Operation

Operation as v_{DS} Is Increased

- *With* v_{DS} is increased and v_{GS} be held constant at a value greater than Vt ;
- MOSFET be operated at a constant overdrive voltage V_{OV};
- v_{DS} appears as a voltage drop across the length of the channel;
- The voltage between the gate and points along the channel decreases from $v_{GS} = V_t + V_{OV}$ at the source end to $v_{GD} = v_{GS} - v_{DS} = V_t + V_{OV} - v_{DS}$ at the drain end.
- Since the channel depth depends on this voltage, and specifically on the amount by which this voltage exceeds Vt , we find that the channel is no longer of uniform depth;

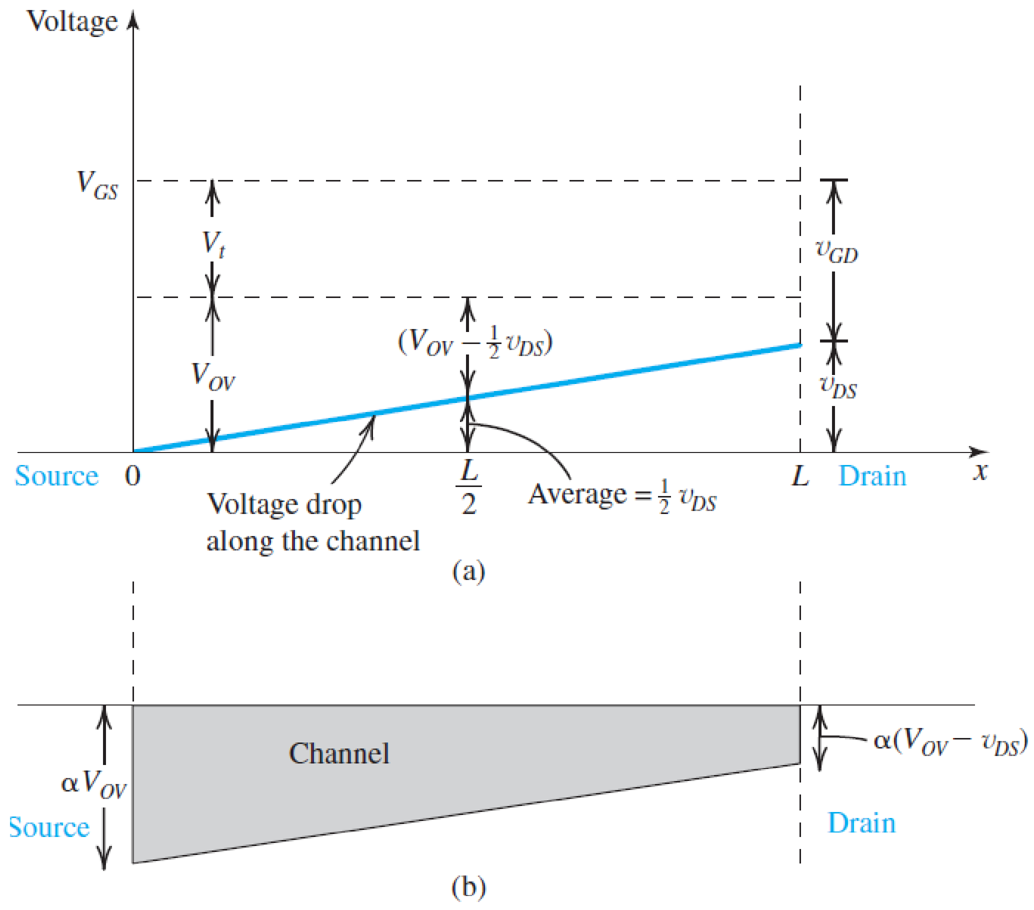


The induced channel acquires a tapered shape, and its resistance increases as v_{DS} is increased. Here, v_{GS} is kept constant at a value > V_t; $v_{GS} = V_t + V_{OV}$.

Transistor MOSFET/Device Structure and Physical Operation

Operation as v_{DS} Is Increased

- As v_{DS} is increased, the channel becomes more tapered and its resistance increases correspondingly.
- The charge in the tapered channel is proportional to the channel cross-sectional area



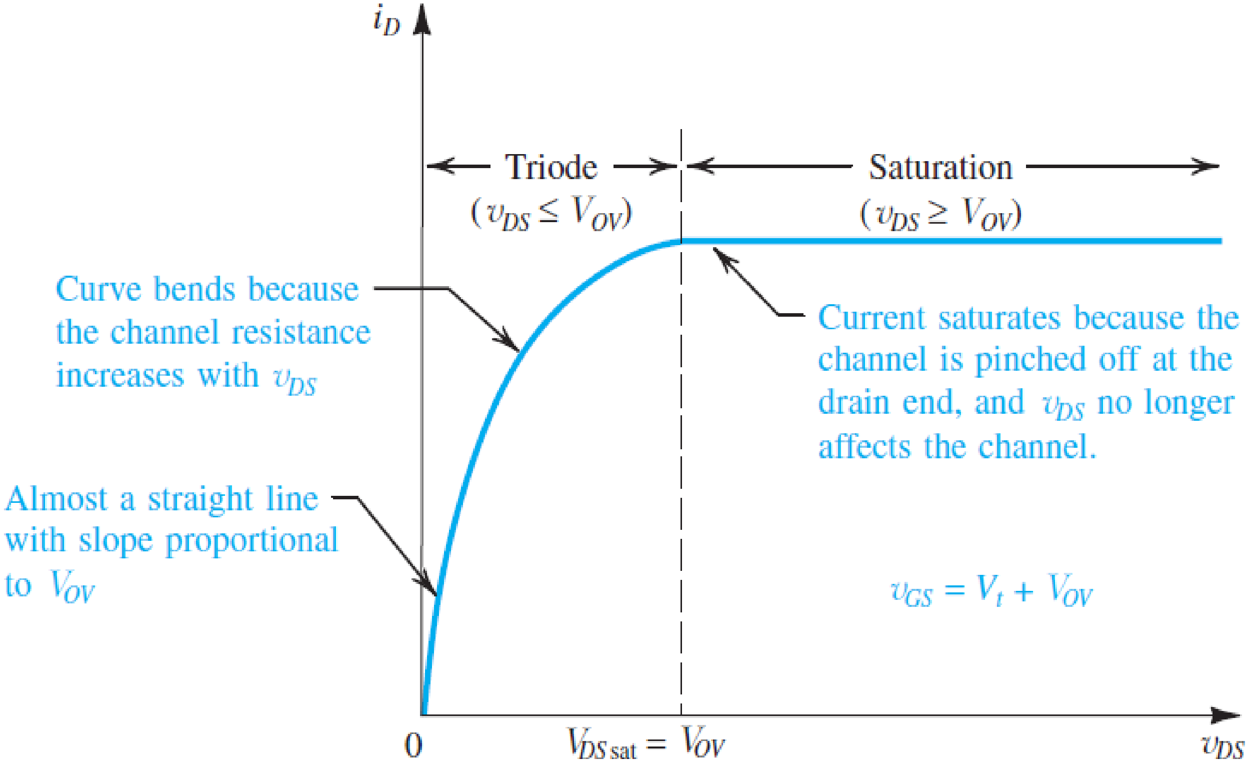
(a) For a MOSFET with $v_{GS}= V_t+V_{OV}$, application of v_{DS} causes the voltage drop along the channel to vary linearly, with an average value of ½ V_{DS} at the midpoint. Since $v_{GD} > V_t$, the channel still exists at the drain end. **(b)** The channel shape corresponding to the situation in **(a)**. While the depth of the channel at the source end is still proportional to V_{OV}, that at the drain end is proportional to $(V_{OV}-v_{DS})$.

Transistor MOSFET/Device Structure and Physical Operation

Operation as v_{DS} Is Increased

- The equation describing this portion of the $i_D - v_{DS}$ curve can be easily derived by utilizing the information in Fig opposite
- the charge in the tapered channel is proportional to the channel cross-sectional area. This area in turn can be easily seen as proportional to

$$\frac{1}{2}[V_{OV}+(V_{OV}-v_{DS})] \text{ or } \left(V_{OV}-\frac{1}{2}v_{DS}\right)$$



- The relationship between i_D and v_{DS} can be found by

$$i_D = k_n'\left(\frac{W}{L}\right)\left(V_{OV}-\frac{1}{2}v_{DS}\right)v_{DS} = k_n'\left(\frac{W}{L}\right)\left(V_{OV}v_{DS}-\frac{1}{2}v_{DS}^2\right)$$

- This relationship describes the semiparabolic portion of the $i_D - v_{DS}$ curve

Transistor MOSFET/Device Structure and Physical Operation

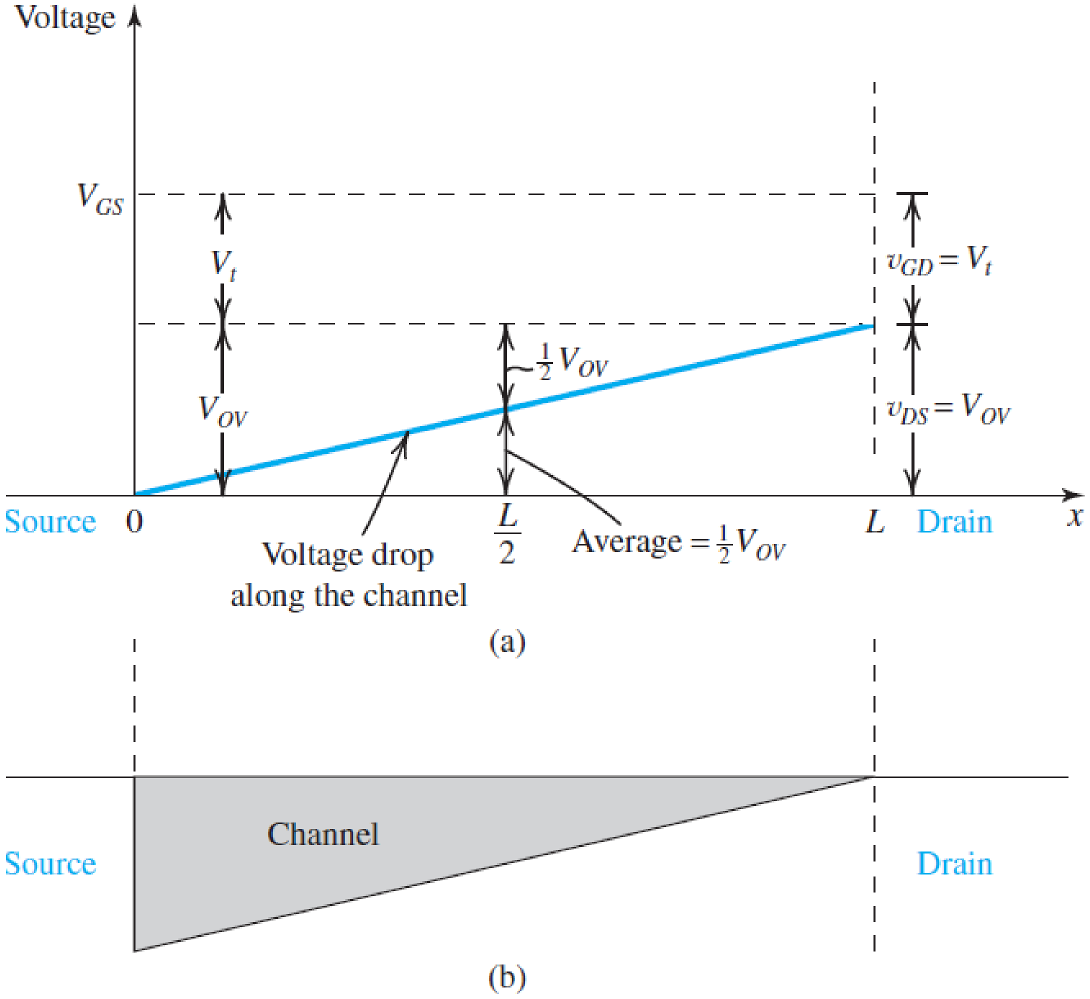
Operation for $v_{DS} \geq V_{OV}$: Channel Pinch-Off and Current Saturation

- for as $v_{DS}=V_{OV}$, $v_{GD}=V_t$, the channel depth at the drain end reduces to zero.

- The zero depth of the channel at the drain end gives rise to the term **channel pinch-off**.

- Increasing v_{DS} beyond this value (i.e., $v_{DS}>V_{OV}$) has no effect on the channel shape and charge, and the current through the channel remains constant at the value reached for $v_{DS}=V_{OV}$. The drain current thus saturates at the value found by

$$i_D = \frac{1}{2} k_n' \left(\frac{W}{L}\right) V_{OV}^2$$



Transistor MOSFET/Device Structure and Physical Operation

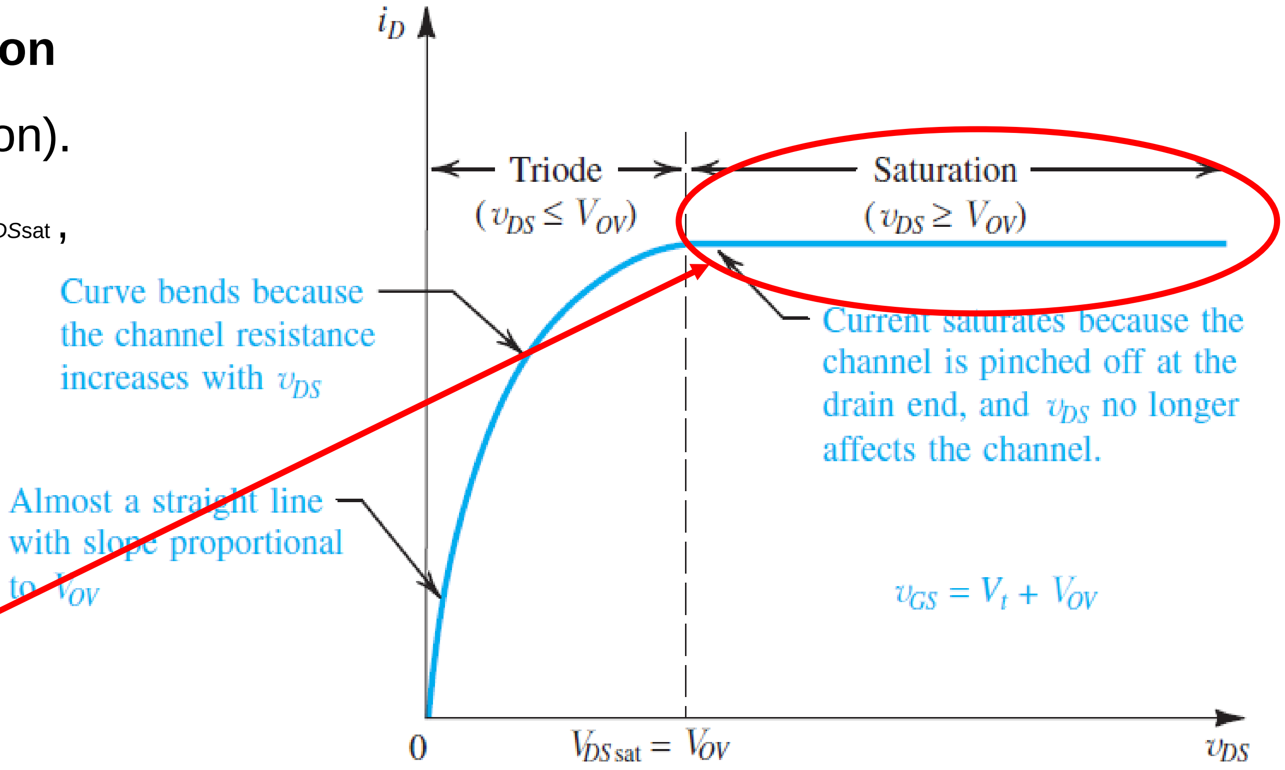
Operation for $v_{DS} \geq V_{OV}$: Channel Pinch-Off and Current Saturation

The MOSFET is then said to have entered the **saturation region** (or, equivalently, the saturation mode of operation).

The voltage v_{DS} at which saturation occurs is denoted V_{DSsat} ,

$$V_{DSsat} = V_{OV} = V_{GS} - V_t$$

$$i_D = \frac{1}{2} k_n' \left(\frac{W}{L} \right) V_{OV}^2$$



FST FES

Transistor MOSFET/Device Structure and Physical Operation

Example 5

Consider a process technology for which $L_{min} = 0.4\ \mu m$, $t_{ox} = 8$ nm, $\mu_n = 450\ cm^2/V \cdot s$, and $V_t = 0.7$ V.

(a) Find C_{ox} and k'_n.

(b) For a MOSFET with $W/L = 8\ \mu m/0.8\ \mu m$, calculate the values of V_{OV}, V_{GS}, and V_{DSmin} needed to operate the transistor in the saturation region with a dc current $I_D = 100\ \mu A$.

(c) For the device in (b), find the values of V_{OV} and V_{GS} required to cause the device to operate as a 1000-Ω resistor for very small v_{DS}.

Solution

(a)

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}} = \frac{3.45 \times 10^{-11}}{8 \times 10^{-9}} = 4.32 \times 10^{-3}\ F/m^2$$

$$= 4.32\ fF/\mu m^2$$

$$k'_n = \mu_n C_{ox} = 450\ (cm^2/V \cdot s) \times 4.32\ (fF/\mu m^2)$$

$$= 450 \times 10^8\ (\mu m^2/V \cdot s) \times 4.32 \times 10^{-15}\ (F/\mu m^2)$$

$$= 194 \times 10^{-6}\ (F/V \cdot s)$$

$$= 194\ \mu A/V^2$$

Transistor MOSFET/Device Structure and Physical Operation

Example 5

(b) For a MOSFET with $W/L = 8\ \mu m/0.8\ \mu m$, calculate the values of V_{OV}, V_{GS}, and $V_{DS\min}$ needed to operate the transistor in the saturation region with a dc current $I_D = 100\ \mu A$.

(b) For operation in the saturation region,

$$i_D = \frac{1}{2} k_n' \frac{W}{L} V_{OV}^2$$

$$100 = \frac{1}{2} \times 194 \times \frac{8}{0.8} V_{OV}^2$$

which results in $V_{OV} = 0.32\ \text{V}$

Thus,

$$V_{GS} = V_t + V_{OV} = 1.02\ \text{V} \quad \text{and} \quad V_{DS\min} = V_{OV} = 0.32\ \text{V}$$

Transistor MOSFET/Device Structure and Physical Operation

Example 5

(c) For the MOSFET in the triode region with v_{DS} very small,

$$r_{DS} = \frac{1}{k_n' \dfrac{W}{L} V_{OV}}$$

Thus

$$1000 = \frac{1}{194 \times 10^{-6} \times 10 \times V_{OV}}$$

which yields

$$V_{OV} = 0.52 \text{ V}$$

Thus,

$$V_{GS} = 1.22 \text{ V}$$

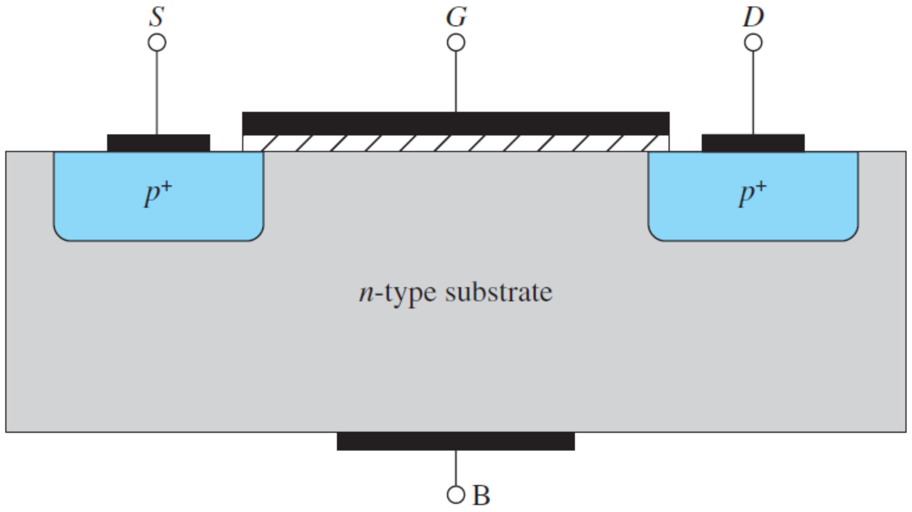
p-Channel MOSFET

- Changing the doping polarities of the substrate and the S/D areas results in a “PMOS” device

- Figure at right shows a cross-sectional view of a *p*-channel enhancement-type MOSFET.

- The structure is similar to that of the NMOS device except that here the substrate is *n* type and the source and the drain regions are *p*+ type;

- The PMOS and NMOS transistors are said to be *complementary* devices.

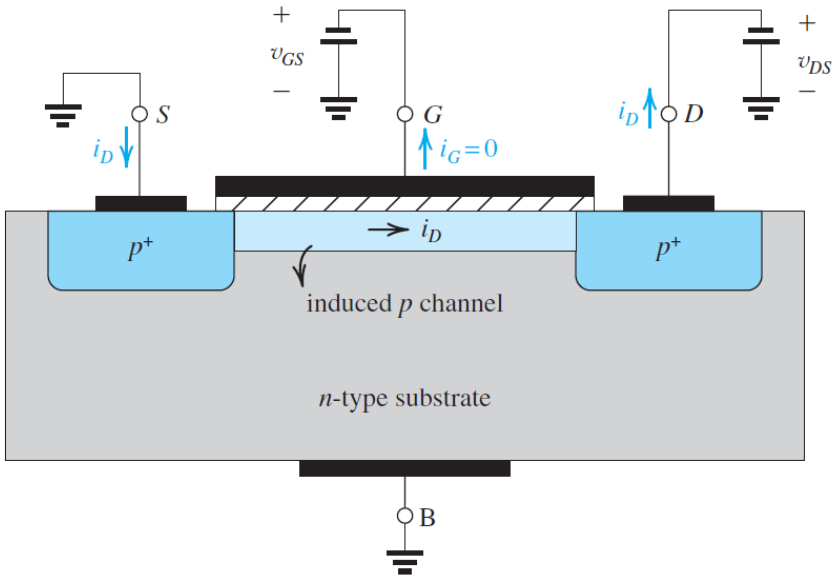


p-Channel MOSFET

- All semiconductor regions are reversed in polarity relative to their counterparts in the NMOS case.
- A negative voltage v_{GS} of magnitude greater than $|Vtp|$ induces a *P* channel, and a negative v_{DS} causes a current i_D to flow from source to drain.

- To induce a channel for current flow between source and drain, a negative voltage is applied to the gate, that is, between gate and source
- By increasing the magnitude of the negative v_{GS} beyond the magnitude of the threshold voltage Vtp, which by convention is negative, a p channel is established as shown in Fig. at right.
- This condition can be described as

$$v_{GS} \leq V_{tp} \qquad |v_{GS}| \geq |V_{tp}|$$



p-Channel MOSFET

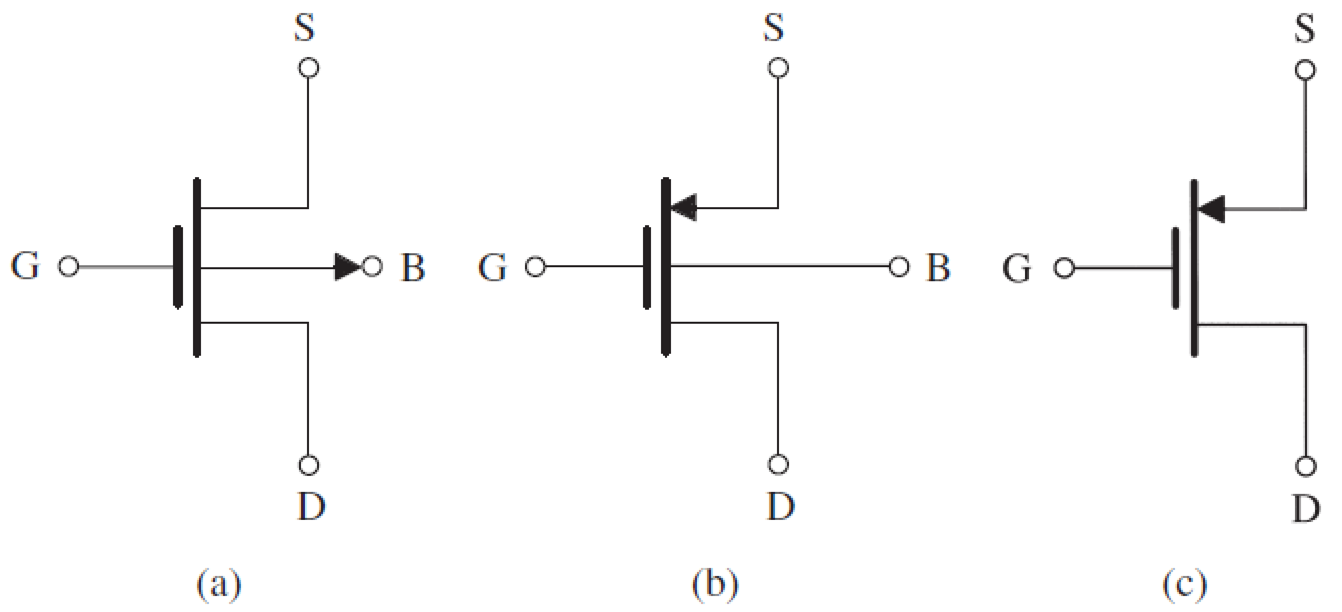
- To cause a current i_D to flow in the p channel, a negative voltage vDS is applied to the drain.
- The current i_D is carried by holes and flows through the channel from source to drain.
- As we have done for the NMOS transistor, we define the process transconductance parameter for the PMOS device as $k'_p = \mu_p C_{ox}$
- where μp is the mobility of the holes in the induced p channel. Typically, μp
- = 0.25μn to 0.5μn and is process-technology dependent.
- The transistor transconductance parameter kp is obtained by multiplying k'_p by the aspect ratio W/L, $k_p = k'_p(W/L)$

- To cause a current i_D to flow in the p channel, a negative voltage vDS is applied to the drain.
- The current i_D is carried by holes and flows through the channel from source to drain.
- As we have done for the NMOS transistor, we define the process transconductance parameter for the PMOS device as
- where μp is the mobility of the holes in the induced p channel. Typically, μp
- = 0.25μn to 0.5μn and is process-technology dependent.
- The transistor transconductance parameter kp is obtained by multiplying k'_p by the aspect ratio W/L,

Characteristics of the p-Channel MOSFET

- The circuit symbol for the p-channel enhancement-type MOSFET is shown in Fig
- in the saturation-region expression for i_D as follows

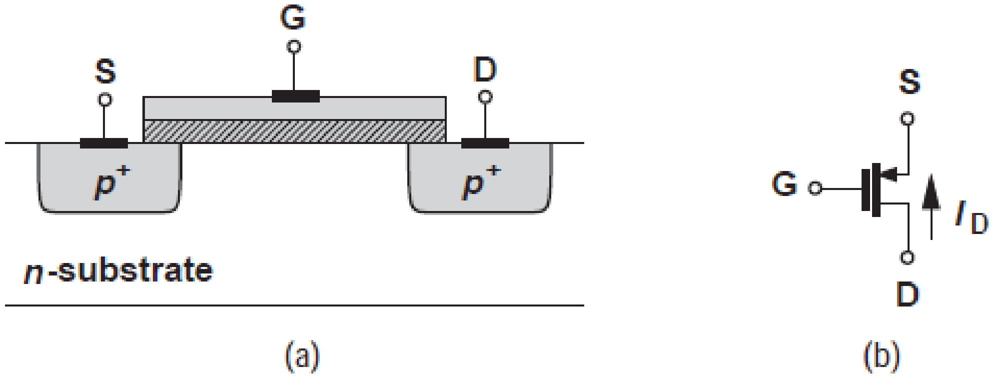
$$i_D = \frac{1}{2}k_p'\left(\frac{W}{L}\right)\left(v_{SG} - |V_{tp}|\right)^2$$

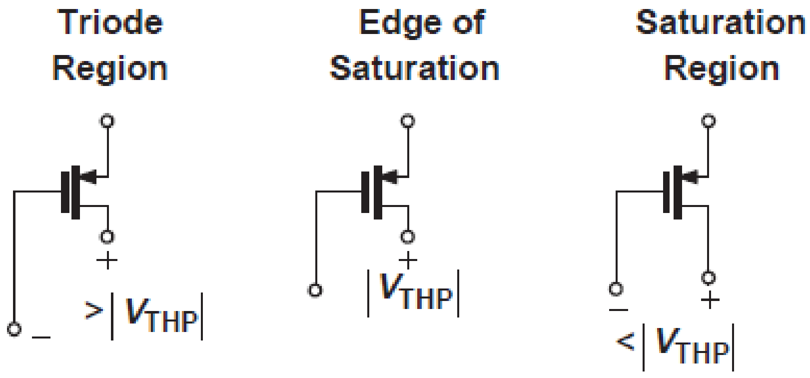


(a) Circuit symbol for the *p*-channel enhancement-type MOSFET. **(b)** Modified symbol with an arrowhead on the source lead. **(c)** Simplified circuit symbol for the case where the source is connected to the body.

Characteristics of the p-Channel MOSFET

- The circuit symbol for the p-channel enhancement-type MOSFET

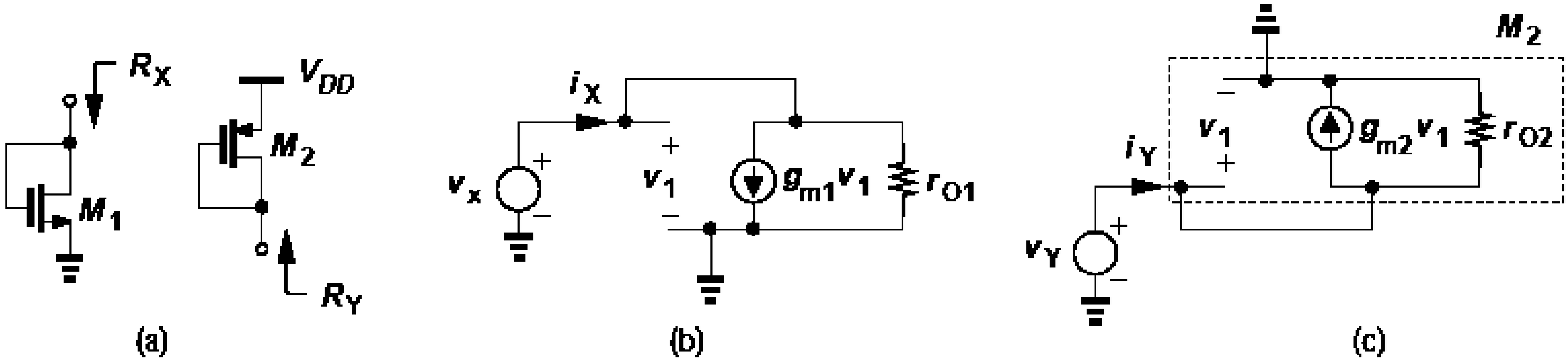




(c)

(a) Structure of PMOS device, (b) PMOS circuit symbol, (c) illustration of triode and saturation regions based on gate and drain voltages.

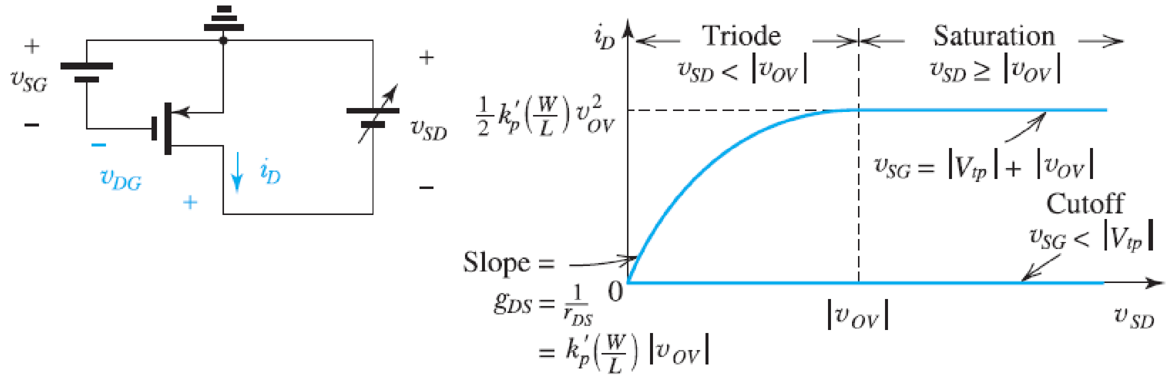
Small signal Model for transistor MOSFET



(a) Diode-connected NMOS and PMOS devices, (b) small-signal model of (a) n-channel, (c) small-signal model of (a) p-channel.

Characteristics of the p-Channel MOSFET

Table 5.2 Regions of Operation of the Enhancement PMOS Transistor



- $v_{SG} < |V_{tp}|$: no channel; transistor in cutoff; $i_D = 0$
- $v_{SG} = |V_{tp}| + |v_{OV}|$: a channel is induced; transistor operates in the triode region or in the saturation region depending on whether the channel is continuous or pinched off at the drain end;

Triode Region

Continuous channel, obtained by:

$$v_{DG} > |V_{tp}|$$

or equivalently

$$v_{SD} < |v_{OV}|$$

Then

$$i_D = k_p'\left(\frac{W}{L}\right)\left[(v_{SG} - |V_{tp}|)v_{SD} - \frac{1}{2}v_{SD}^2\right]$$

or equivalently

$$i_D = k_p'\left(\frac{W}{L}\right)\left(|v_{OV}| - \frac{1}{2}v_{SD}\right)v_{SD}$$

Saturation Region

Pinched-off channel, obtained by:

$$v_{DG} \leq |V_{tp}|$$

or equivalently

$$v_{SD} \geq |v_{OV}|$$

Then

$$i_D = \frac{1}{2}k_p'\left(\frac{W}{L}\right)(v_{SG} - |V_{tp}|)^2$$

or equivalently

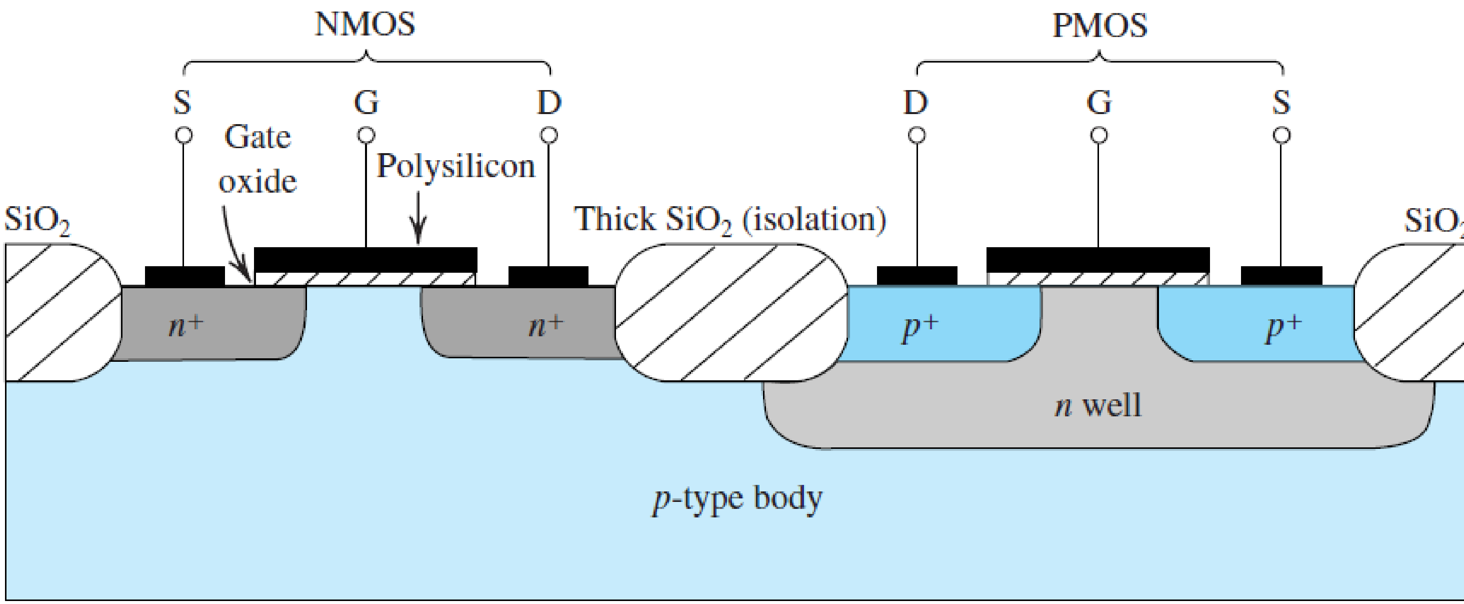
$$i_D = \frac{1}{2}k_p'\left(\frac{W}{L}\right)v_{OV}^2$$

Complementary MOS or CMOS

- As the name implies, complementary MOS technology employs MOS transistors of both polarities. Although CMOS circuits are somewhat more difficult to fabricate than NMOS, the availability of complementary devices makes possible many powerful circuit configurations.

- At the present time CMOS is the most widely used of all the IC technologies.and that for both analog and digital circuits.

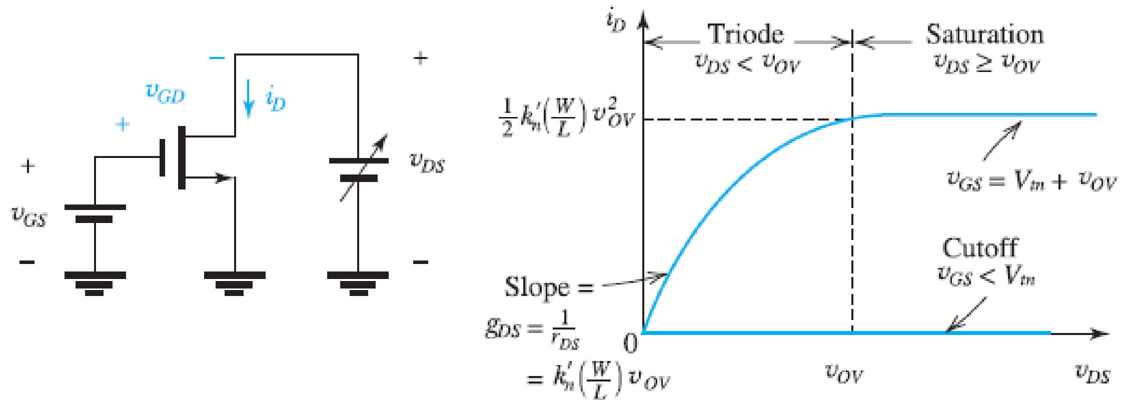
- CMOS technology has virtually replaced designs based on NMOS transistors alone.



Cross section of a CMOS integrated circuit. Note that the PMOS transistor is formed in a separate *n*-type region, known as an *n* well. Another arrangement is also possible in which an *n*-type substrate (body) is used and the *n* device is formed in a *p* well.

n-channel Current–Voltage Characteristics/i_D–v_{DS} Characteristics

- The summary at right provides a compilation of the conditions and the formulas for the operation of the NMOS transistor in each of the three possible regions: the cutoff region, the triode region, and the saturation region. The first two are useful if the MOSFET is to be utilized as a switch.
- To operate in the triode region, the gate voltage must exceed the drain voltage by at least Vtn volts, which ensures that the channel remains continuous (not pinched off).
- To operate in saturation, the channel must be pinched off at the drain end; pinch-off is achieved here by keeping v_D higher than $v_G - V_{tn}$,that is, not allowing v_D to fall below v_G by more than Vtn volts.
- If the MOSFET is to be used to design an amplifier, it must be operated in the saturation region.



- $v_{GS} < V_{tn}$: no channel; transistor in cutoff; $i_D = 0$
- $v_{GS} = V_{tn} + v_{OV}$: a channel is induced; transistor operates in the triode region or the saturation region depending on whether the channel is continuous or pinched off at the drain end;

Triode Region

Continuous channel, obtained by:

$$v_{GD} > V_{tn}$$

or equivalently:

$$v_{DS} < v_{OV}$$

Then,

$$i_D = k_n'\left(\frac{W}{L}\right)\left[(v_{GS} - V_{tn})v_{DS} - \frac{1}{2}v_{DS}^2\right]$$

or equivalently,

$$i_D = k_n'\left(\frac{W}{L}\right)\left(v_{OV} - \frac{1}{2}v_{DS}\right)v_{DS}$$

Saturation Region

Pinched-off channel, obtained by:

$$v_{GD} \leq V_{tn}$$

or equivalently:

$$v_{DS} \geq v_{OV}$$

Then

$$i_D = \frac{1}{2}k_n'\left(\frac{W}{L}\right)(v_{GS} - V_{tn})^2$$

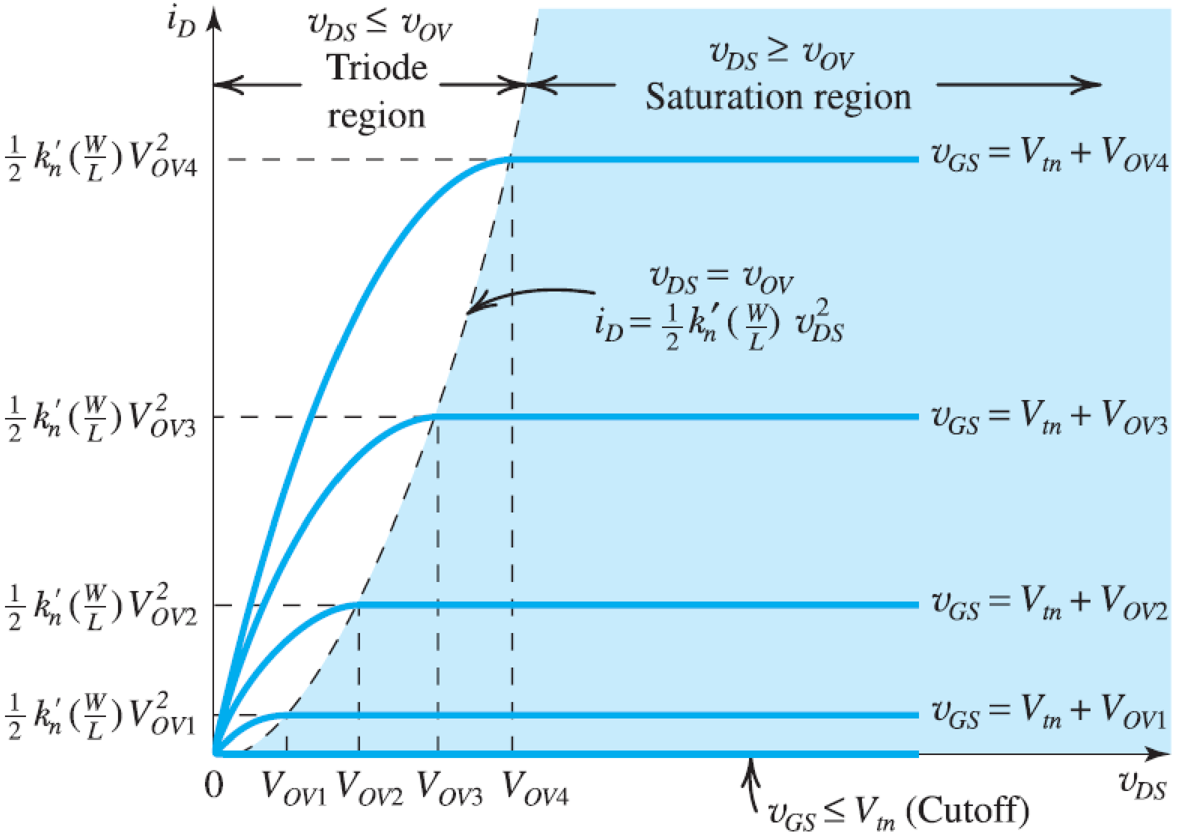
or equivalently,

$$i_D = \frac{1}{2}k_n'\left(\frac{W}{L}\right)v_{OV}^2$$

n-channel Current–Voltage Characteristics

- A set of iD −vDS characteristics for the NMOS transistor is shown in Fig. at right To operate in the triode region, the gate voltage must exceed the drain voltage by at least Vtn volts, which ensures that the channel remains continuous (not pinched off).

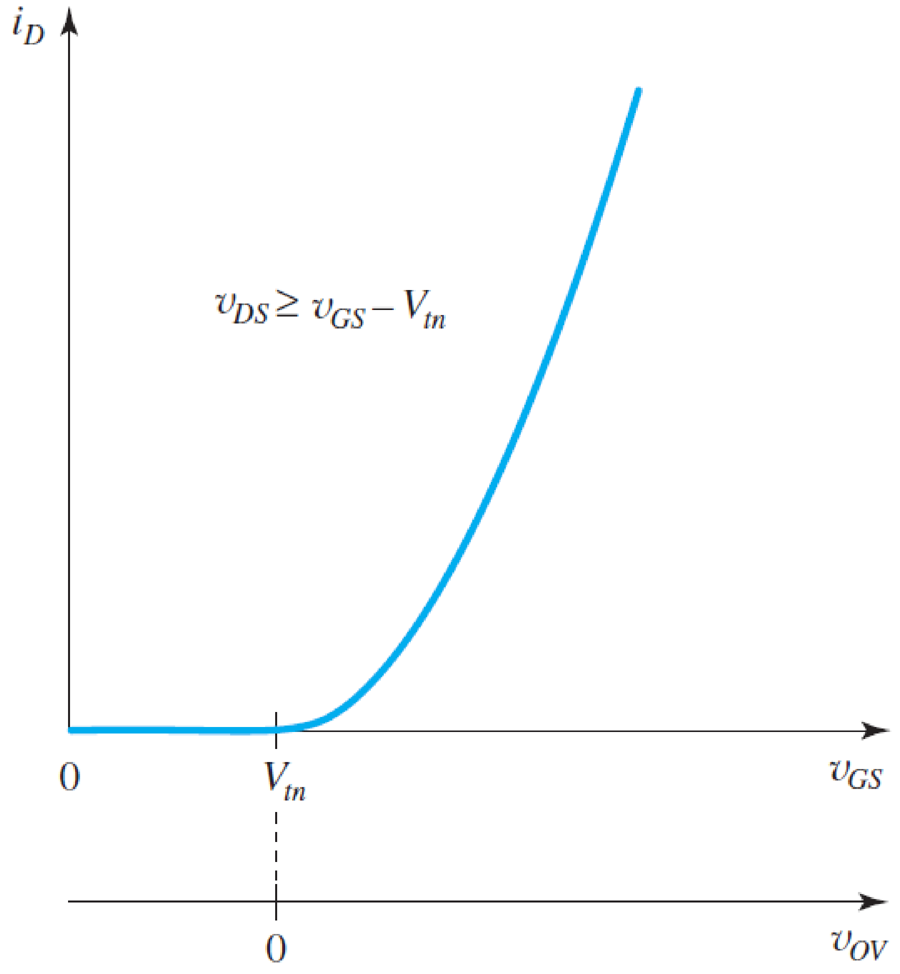
- Each graph is obtained by setting v_{GS} above Vtn by a specific value of overdrive voltage, denoted V_{OV1}, V_{OV2}, V_{OV3}, and V_{OV4}.



n-channel i_D–v_{GS} Characteristic

- When the MOSFET is used to design an amplifier, it is operated in the saturation region.
- in saturation the drain current is constant determined by v_{GS} (or v_{OV}) and is independent of v_{DS}.
- In effect, then, the MOSFET operates as a voltage-controlled current source with the control relationship described by

$$i_D = \frac{1}{2}k_n'\left(\frac{W}{L}\right)(v_{GS} - V_{tn})^2 \quad = \frac{1}{2}k_n'\left(\frac{W}{L}\right)v_{OV}^2$$

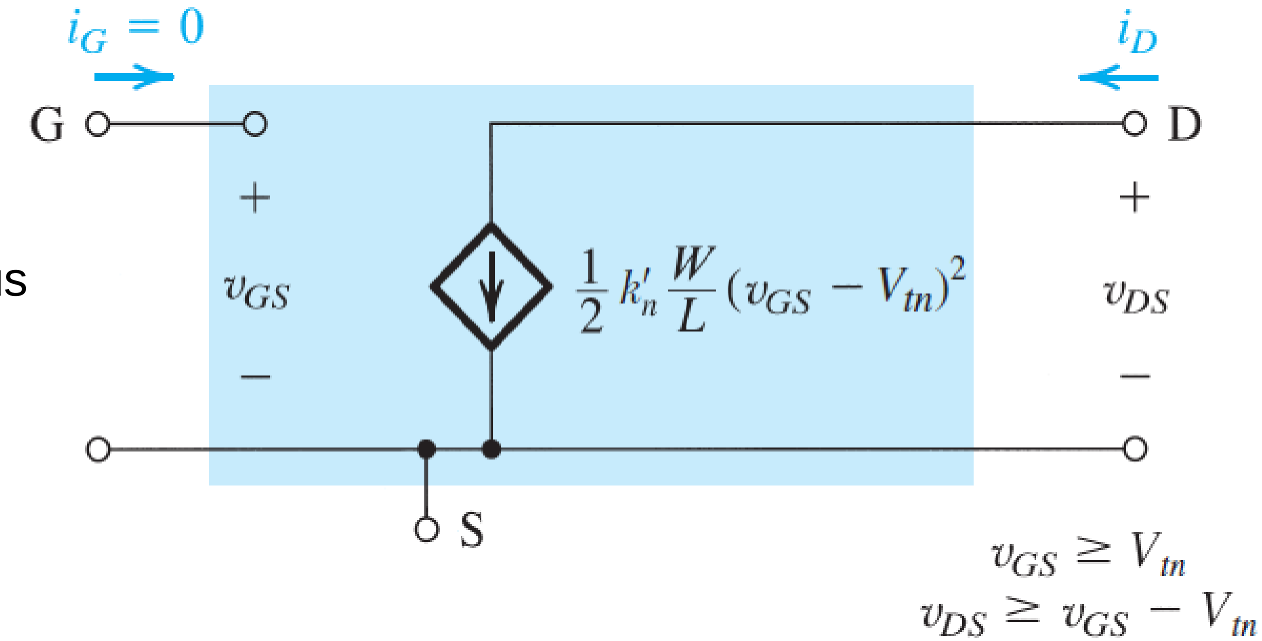


NMOSFET as a current source

- The view of the MOSFET in the saturation region as a voltage-controlled current source is illustrated by the equivalent-circuit representation shown in Fig.

- The current source is ideal, with an infinite output resistance representing the independence, in saturation, of i_D from v_{DS}
- This, the idealized model of device operation utilized thus far.

MOSFET Ideal Model

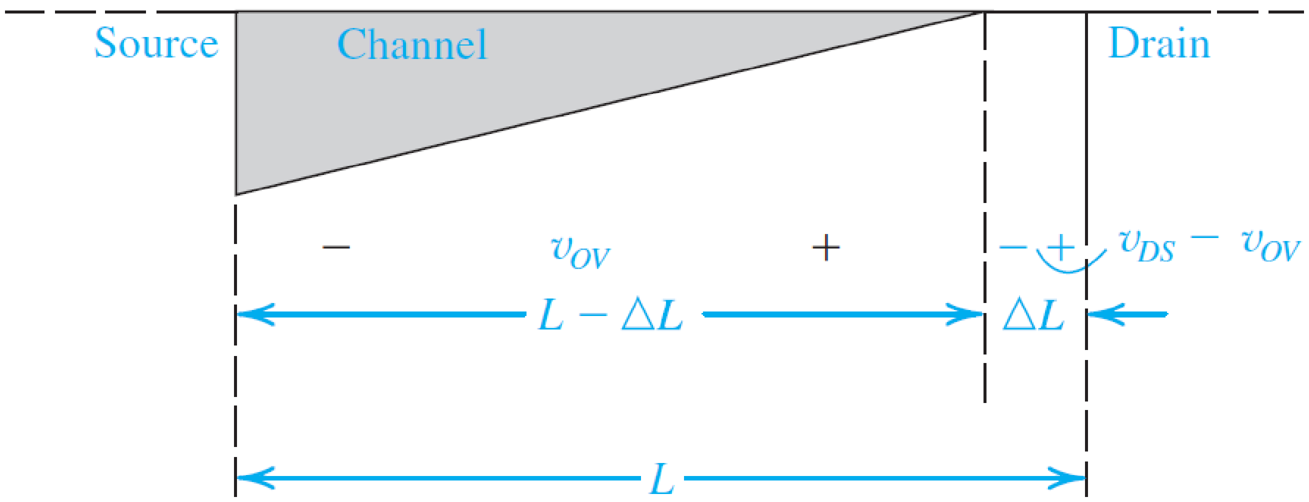


Large-signal, equivalent-circuit model of an *n*-channel MOSFET operating in the saturation region.

Finite Output Resistance in Saturation Region

- In practice, increasing v_{DS} beyond v_{OV} does affect the channel somewhat. Specifically, as v_{DS} is increased, the channel pinch-off point is moved slightly away from the drain, toward the source.

- This is illustrated in Fig. below, from which we note that the voltage across the channel remains constant at v_{OV} , and the additional voltage applied to the drain appears as a voltage drop across the narrow depletion region between the end of the channel and the drain region.



Increasing v_{DS} beyond v_{DSsat} causes the channel pinch-off point to move slightly away from the drain, thus reducing the effective channel length (by ΔL).

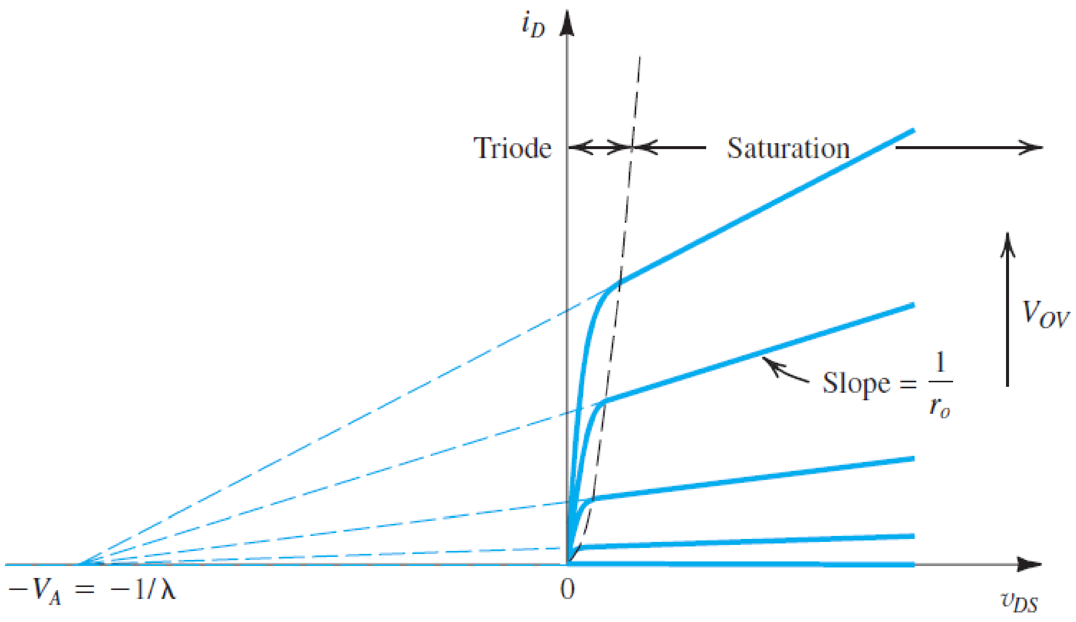
Finite Output Resistance in Saturation Region

- This voltage accelerates the electrons that reach the drain end of the channel and sweeps them across the depletion region into the drain.
- the channel length is in effect reduced, from L to $L - \Delta L$, a phenomenon known as **channel-length modulation**.
- This effect can be accounted for in the expression for i_D by including a factor $1+\lambda(v_{DS}-v_{OV})$ or, for simplicity, $(1+\lambda v_{DS})$,

$$i_D = \frac{1}{2}k_n'\left(\frac{W}{L}\right)(v_{GS} - V_{tn})^2(1+\lambda v_{DS})$$

- With $V_A = \frac{1}{\lambda}$ and $V_A = V_A' L$

The voltage VA is usually referred to as the Early voltage (range of 5 V/µm to 50V/µm)



Effect of v_{DS} on i_D in the saturation region. The MOSFET parameter V_A depends on the process technology and, for a given process, is proportional to the channel length L.

Finite Output Resistance in Saturation Region

The Equation

$$i_D = \frac{1}{2}k_n'\left(\frac{W}{L}\right)(v_{GS} - V_{tn})^2(1+\lambda v_{DS})$$

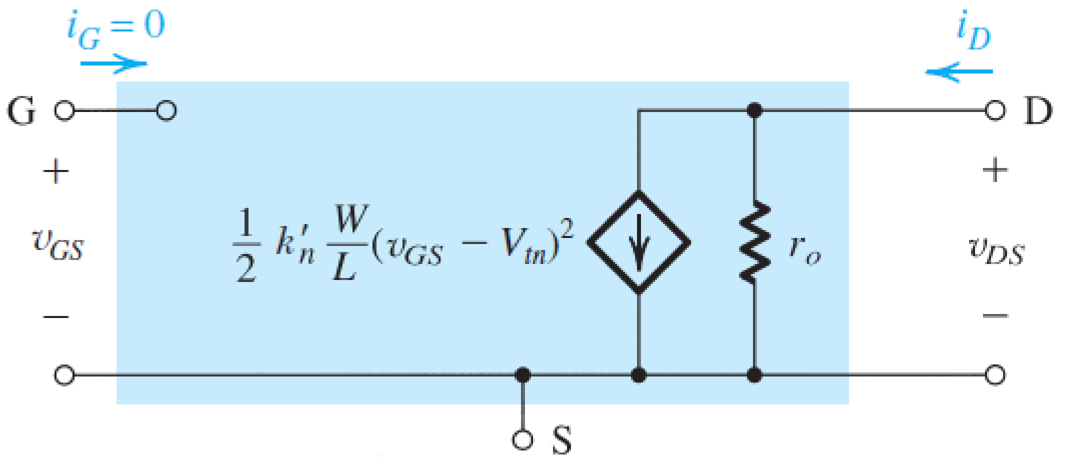
indicates that when channel-length modulation is taken into account, the saturation values of i_D depend on v_{DS}. Thus, for a given v_{GS}, a change v_{DS} yields a corresponding change i_D in the drain current i_D.

It follows that the output resistance of the current source representing i_D in saturation is no longer infinite. Defining the outp

$$r_o \equiv \left[\frac{\partial i_D}{\partial v_{DS}}\right]^{-1}_{v_{GS}\ \text{constant}} = \left[\lambda \frac{k_n'}{2}\frac{W}{L}(V_{GS}-V_{tn})^2\right]^{-1} = \frac{1}{\lambda I_D} = \frac{V_A}{I_D}$$

where I_D is the dr ... annel-length modulation taken into accou

$$I_D = \frac{1}{2}k_n'\frac{W}{L}(V_{GS}-V_{tn})^2$$



Large-signal, equivalent-circuit model of the *n*-channel MOSFET in saturation, incorporating
the output resistance r_o.

Finite Output Resistance in Saturation Region for p-channel MOSFET-

With channel length modulation, in the saturation region the current expressed as

$$i_D = \frac{1}{2} k_p' \left(\frac{W}{L}\right) \left(v_{SG} - |V_{tp}|\right)^2 (1 + |\lambda| v_{SD})$$

And the finite resistance is expressed as

$$r_o \equiv \left[\frac{\partial i_D}{\partial v_{DS}}\right]^{-1}_{v_{GS}\ \text{constant}} = \left[\lambda \frac{k_n'}{2} \frac{W}{L} (V_{GS} - V_{tn})^2\right]^{-1} = \frac{1}{\lambda I_D} = \frac{V_A}{I_D}$$

$$I_D = \frac{1}{2} k_n' \frac{W}{L} (V_{GS} - V_{tn})^2$$